Collective Trading Structure and Market Risk: From Aggregate Observations to Dynamic Inference

Econophysics Risk Lab contributors

Collective dynamics, inference and market risk

November 6, 2026

Abstract

Can aggregate trading observations reveal collective structure that helps explain and assess market risk? We connect exact finite-network moments, effective trading weights and an interdependent decision network to a common observable-to-risk framework. Finite-network inversion and Bitcoin execution diagnostics distinguish identifiable statistical structure from unobserved investor identity. A calendar-time extension now separates persistent trading pressure, recovery of predictable price impact and the environment receiving collective trades. Controlled experiments show that common decision innovations increase one-hour risk more strongly at lower simulated shock capacity: the paired increment falls from 3.31×10^{-6} to 1.64×10^{-6} across the tested capacity range. Independent future replications show that these effects contain genuine state variation, yet their more precise recovery does not improve risk prediction beyond existing activity controls. Hidden activity, size, liquidity and weight-concentration states jointly add information that the tested macro readouts do not fully recover. A corresponding past-only Bitcoin closure produces an internal QLIKE point improvement of 5.79% over a same-field HAR reference, with uncertainty spanning zero. An exact limiting-case cancellation and direct-moment controls show why this improvement cannot be assigned uniquely to coordination inference. The contribution is an executable account of when collective trading transmits risk, what macro observations reveal, and which additional predictive claims require evidence beyond a positive average mechanism effect.

1 Introduction

Aggregate price and trading records compress many interacting decisions into returns, executed volume, counts and signed aggressive flow. A large decision can be executed in pieces; many independently acting traders can share a signal; and the same pressure can move prices differently as accommodation changes. The central question is whether a small observable interface can reveal risk-relevant collective structure, rather than merely reproduce a familiar market pattern.

The Cont–Bouchaud model links coordinated group actions to aggregate fluctuations[1]. Adaptive market models add switching and feedback[2], while calibration research distinguishes observable resemblance from parameter identification[6, 7]. We use these distinctions to connect a tractable finite-network reference to a weighted fast–slow decision model and an explicit observable-state inference task. A transaction price records an exchange under current information and trading constraints; our short-horizon model studies pressure and accommodation, not a universal intrinsic-value or eventual price-reversion law.

The research follows the sequence *macro observations → effective statistical structure → future joint trading/price paths → risk evaluation and application.* Exact moments and finite-ER inversion establish the initial inferential route. Weighted covariance and execution checks define what can be carried into market observations. The same interdependent network then supplies state dynamics, local perturbation response and common-signal/feedback experiments. Observable-only inference and common-path risk test the reverse route without supplying the target's true state or future.

Four questions organize the evidence: which structural quantities the reference identifies; which execution and temporal observations distinguish them; how the retained dynamic law turns state uncertainty into future risk; and whether that information improves an equally supplied forecast or

reduces total measurement cost. Controlled simulations, Bitcoin observations and sector-portfolio applications have distinct roles. The completed experiments provide conditional mechanism results, numerical propagation checks and prediction comparisons; market observation calibration and independent predictive confirmation are evaluated as separate requirements.

The most recent continuation asks why a positive coordination effect may add little forecast value. Independent future repetitions separate genuine state heterogeneity from label noise. A paired capacity–correlation–feedback grid then quantifies the environments in which coordination raises risk. Finally, source-state information tests and a matched-field Bitcoin closure connect that explanation back to the observable interface. Their combined result is an environment-conditioned risk mechanism with a precise representation boundary: an interpretable state calculation is not automatically a uniquely informative forecast.

2 Finite network model and observable moments

2.1 Agents, actions, and the observation interface

Each period draws $H \sim G(N,p)$, where $p = \rho/(N-1)$, $N \geq 2$, and $0 \leq \rho \leq N-1$. Nodes are model traders, edges represent coordination, and a connected cluster C contains s_C traders. Each cluster shares $\xi_C \in \{-1,0,1\}$ with probabilities $q/2, 1-q, q/2$. Actions are independent across clusters and independent of the graph. Throughout, q is the total activity probability. The implementation uses the equivalent per-direction probability $a = q/2$.

Each active trader submits a unit order $u > 0$. A simplified external counterparty absorbs all submitted orders, and the log-price response is linear. The observations are

$$r = \frac{\lambda}{N}\sum_C s_C \xi_C, \qquad V = u\sum_C s_C|\xi_C|, \tag{1}$$
$$S_j = \sum_C s_C^j, \qquad \lambda > 0, \quad 0 < q < 1.$$

Signed price effects can offset, whereas trading activity adds. Here V counts executed activity and λ is a model response scale. Table 1 defines the notation, and Fig. 1 illustrates the observation rule.

Table 1. Model notation. A connected group shares one action within a period.

Symbol	Definition
N, ρ	Number of traders and expected ER degree, with $p = \rho/(N-1)$
C, s_C, S_j	Connected group, its size, and the power sum $S_j = \sum_C s_C^j$
q, ξ_C	Total activity probability and shared sell/inactive/buy action
u, λ	Unit order size and positive log-price response scale
r, V	Log return and total executed model volume per period
$A_N(\rho)$	Expected squared-size sum $\mathbb{E}S_2$ under the finite ER graph law

2.2 Exact moments and their mechanistic use

Zero-mean cluster actions eliminate cross terms in return variance. Conditional mean volume does not depend on the graph. Writing $A = \mathbb{E}S_2$ gives

$$\mathbb{E}V = quN, \tag{2}$$
$$\mathrm{Var}(r) = q(\lambda/N)^2 A,$$
$$\mathrm{Var}(V) = q(1-q)u^2 A.$$

Equal mean volume therefore need not imply equal risk. At fixed N, q, u, λ, a larger sum of squared cluster sizes increases return variance. For 100 independent traders, $S_2 = 100$; ten clusters of ten traders give $S_2 = 1000$. Variance differs tenfold in this model example while mean volume remains unchanged.

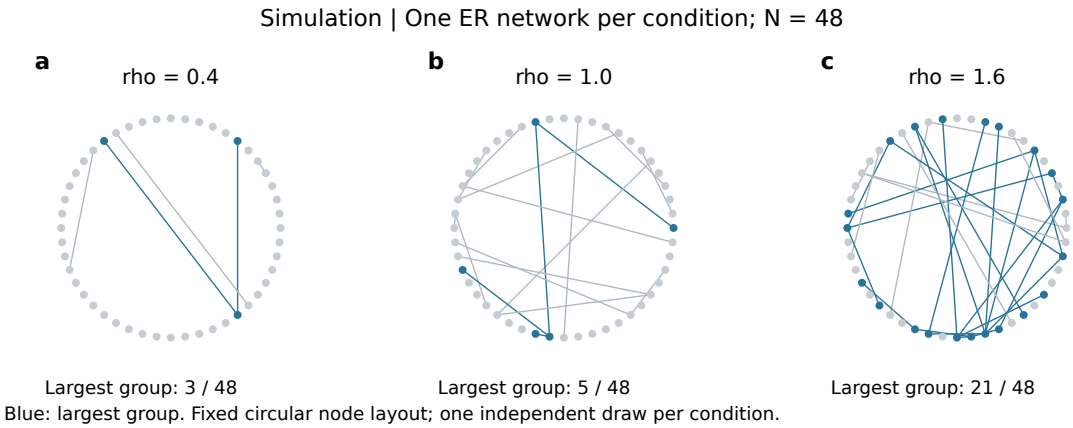


Figure 1. Shared group actions generate net price effects and total trading activity through the same execution rule. Nodes are model traders and connected groups coordinate their actions. [F01]

Higher-order relations retain the connection between collective activity and large price changes:

$$\begin{aligned} \mathrm{Cov}(r^2, V) &= u(\lambda/N)^2 q(1-q)\mathbb{E}S_3, \\ \kappa_3(V) &= u^3 q(1-q)(1-2q)\mathbb{E}S_3. \end{aligned} \tag{3}$$

Signed r and V have zero population covariance, while $|r|$ and V are positively associated under this interface. Price direction and magnitude thus require separate analysis. The empirical analysis below tests stronger joint restrictions than this association alone.

Conditional on the graph, the joint characteristic function is

$$\mathbb{E}[e^{itr+izV} \mid H] = \prod_C [1 - q + qe^{izus_C} \cos(t\lambda s_C/N)]. \tag{4}$$

It depends only on the multiset of cluster sizes, directly supporting the identification boundary below. For fixed N, $|r| \le \lambda$, so the mathematical tail is bounded. This bounded support sets the domain for the finite-range tail tests below.

Source tables and methods: `docs/theory/FOUNDATIONS.md`.

3 Percolation and finite return distributions

In the infinite sparse ER limit, degrees are Poisson(ρ) and $G_0(z) = G_1(z) = e^{\rho(z-1)}$. The edge-exploration generating function satisfies $H_1(z) = zG_1(H_1(z))$. A giant component emerges when the mean branching number crosses one. Its fraction satisfies

$$P_\infty = 1 - e^{-\rho P_\infty}, \qquad \rho_c = 1. \tag{5}$$

Below threshold, the expected cluster size of a uniformly selected node is $1/(1-\rho)$. In finite systems, S_2/N differs from the statistic excluding one largest component,

$$\chi_f = \frac{\sum_{C \neq C_{\max}} s_C^2}{\sum_{C \neq C_{\max}} s_C} \tag{6}$$

which is recorded as missing when no finite component remains.

We scanned $N = 128, 256, 512$, nine ρ values, and 40 graphs per cell, totaling 1,080 graphs. Figure 2 shows a smooth finite-size connectivity transition. Grid resolution and fluctuations affect peak positions. The infinite-system threshold organizes the finite-size comparison; the measured peak also depends on system size and grid resolution.

The mean-field critical law $n_s \propto s^{-5/2}$ concerns cluster number density. Selecting a node uniformly instead weights cluster sizes by s. Returns sum signed cluster actions; activity, cancellation, and bounded support affect their tails. For density $f(x) \propto x^{-\alpha}$, the complementary cumulative

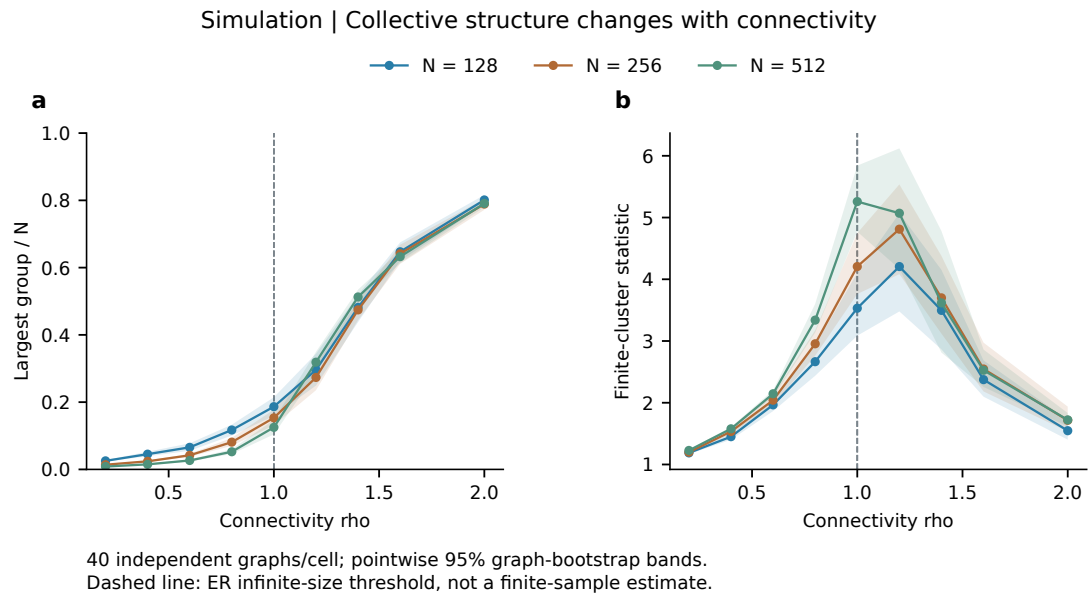


Figure 2. Finite-size connectivity transition for three population sizes. Each N/ρ cell contains 40 independent graphs. The vertical line marks the infinite sparse-ER threshold; finite-system peaks depend on size and grid resolution. Source: simulation_study_v1/phase_summary.csv. [F03]

distribution (CCDF) exponent is $\alpha - 1$. These different sampling and aggregation operations require separate exponent definitions.

Five conditions each generated 8,192 periods, totaling 40,960 simulated intervals. Over the prespecified absolute net-order range $m = 4, \ldots, 128$, all three ρ conditions rejected the simple finite-support pure power law. The raw $p = 0.005$ is the resolution floor of 199 parametric-bootstrap draws; Holm adjustment across three tests gives $p = 0.015$. A better relative AIC under some conditions does not imply adequate absolute fit. The implementation follows maximum-likelihood, Kolmogorov–Smirnov (KS), and alternative-distribution principles[4], adapted to finite integer support.

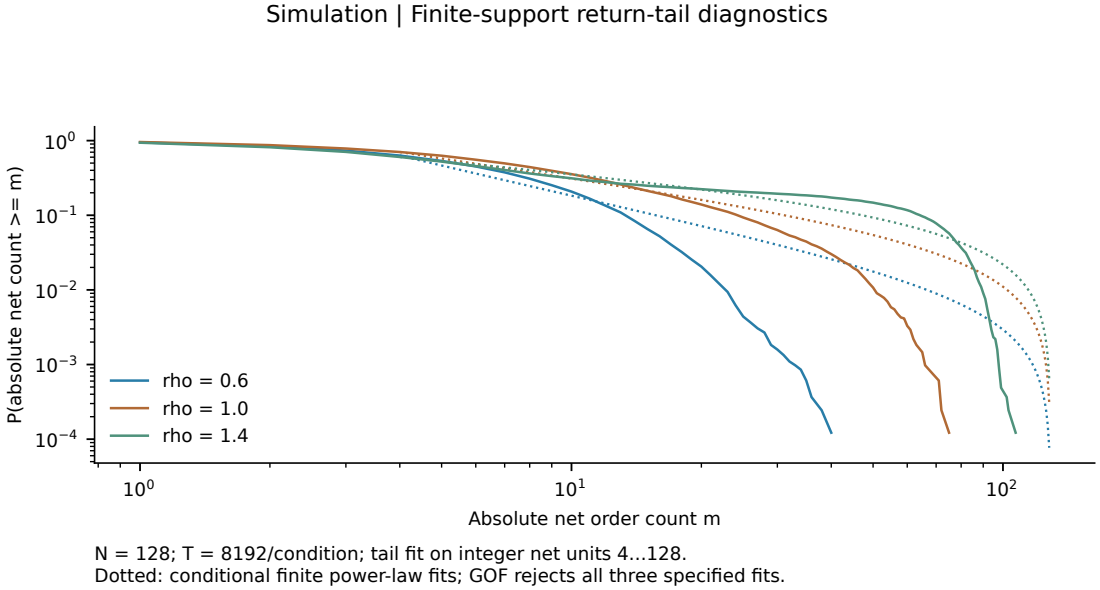


Figure 3. Return CCDFs and finite-support tail fits over the stated net-order range. The pure-power-law goodness-of-fit test rejects all three fitted connectivity conditions; relative AIC comparisons measure a separate property. [F05]

Collective amplification therefore survives even where this finite return distribution fails a pure-power-law fit. Cluster geometry and return tails answer different questions. Appendix B compares cluster distributions, population-size changes, and temporal aggregation.

4 Parameter recovery and observational equivalence

4.1 Finite-graph moment inversion

For known N, u under the fixed observation assumptions, population moments give

$$q = \frac{\mathbb{E}V}{uN}, \qquad A_N(\rho) = \frac{\mathrm{Var}(V)}{q(1-q)u^2}, \tag{7}$$
$$\lambda^2 = N^2u^2(1-q)\frac{\mathrm{Var}(r)}{\mathrm{Var}(V)}.$$

The ER mapping $A_N(\rho) = \mathbb{E}S_2$ is strictly increasing and therefore supports inversion. Let $C_s(p)$ be the probability that $G(s,p)$ is connected, and let $\pi_{N,s}$ be the probability that a uniformly selected trader belongs to a group of size s.

At fixed p, Gilbert's connectivity recursion is[5]:

$$C_s = 1 - \sum_{k=1}^{s-1} \binom{s-1}{k-1} C_k (1-p)^{k(s-k)},$$
$$\pi_{N,s} = \binom{N-1}{s-1} C_s (1-p)^{s(N-s)}, \tag{8}$$
$$A_N(\rho) = N \sum_s s\pi_{N,s}, \qquad C_1 = 1.$$

This classical recursion supplies the finite-size map needed by the moment estimator. The identity $A_N = N + N(N-1)\Pr(1 \leftrightarrow 2)$ connects it to pairwise connectivity. Coupling graphs by adding edges makes this probability strictly increasing in p for $N \geq 2$, which gives the population inverse its uniqueness. The implementation used 380 decimal digits, checked against higher precision and small-graph enumeration.

Ten independently regenerated studies produced 4,920 windows of 256 periods, totaling 1,259,520 simulated intervals. Each study used 252 training, 84 validation, and 84 test windows, plus 72 unseen intermediate-parameter windows. Moment inversion and classical radial-basis-function (RBF) methods share evaluation units. The return-only view contains return standard deviation, mean absolute return, the 95th and 99th absolute-return quantiles, and the zero-return fraction. The volume-only view contains volume mean, standard deviation and 95th percentile. Joint inputs combine these views and the Pearson correlation of volume with absolute return. Seeds and label metadata are excluded.

The supervised reference is kernel ridge regression with a Gaussian radial-basis kernel. For standardized observed features x, training labels $y_i = \rho_i$, and training mean $\bar{y}$,

$$K_{ij} = \exp[-\gamma\|x_i - x_j\|^2],$$
$$\widehat{\rho}(x) = \bar{y} + k(x)^\top (K + \alpha_K I)^{-1}(y - \bar{y}\mathbf{1}). \tag{9}$$

Here $k_i(x) = \exp[-\gamma\|x - x_i\|^2]$ and α_K is the ridge penalty. Median imputation, missingness indicators and standardization are fitted on training windows only. Validation MAE selects the penalty and kernel width from the fixed configuration grid. For positive return standard deviation, the scale-invariant view divides the three return-magnitude summaries by that scale and retains the zero fraction and volume summaries; degenerate windows follow the imputation rule. Table 2 and Fig. 4 report the matched-scale results.

On the original grid, joint RBF reduced average MAE by 44.49% relative to return-only RBF. Volume-only RBF performed better than joint RBF in all ten studies. For unseen parameters, moment inversion achieved MAE 0.027114, compared with 0.026882 for joint RBF. Mechanistic formulas can therefore provide interpretable estimators within the specified family.

4.2 Price rescaling separates representation error from information loss

Each window undergoes $r \mapsto br$, with $b = 0.5, 1, 2$, while V remains unchanged. These are paired transformations of the same windows. Doubling the price response increases raw joint-RBF MAE

Table 2. Connectivity recovery in the specified ER family: mean MAE over ten independent studies, in units of ρ.

Method	Original grid	Unseen intermediate ρ
Return-only RBF	0.048660	0.045077
Joint RBF	0.027013	0.026882
Volume-only RBF	0.022196	0.036849
Scale-invariant joint RBF	0.032106	0.031013
Moment inversion	0.028372	0.027114

Each study regenerates training, validation, and evaluation windows. Source: depth_study_v1/summary_metrics.csv.

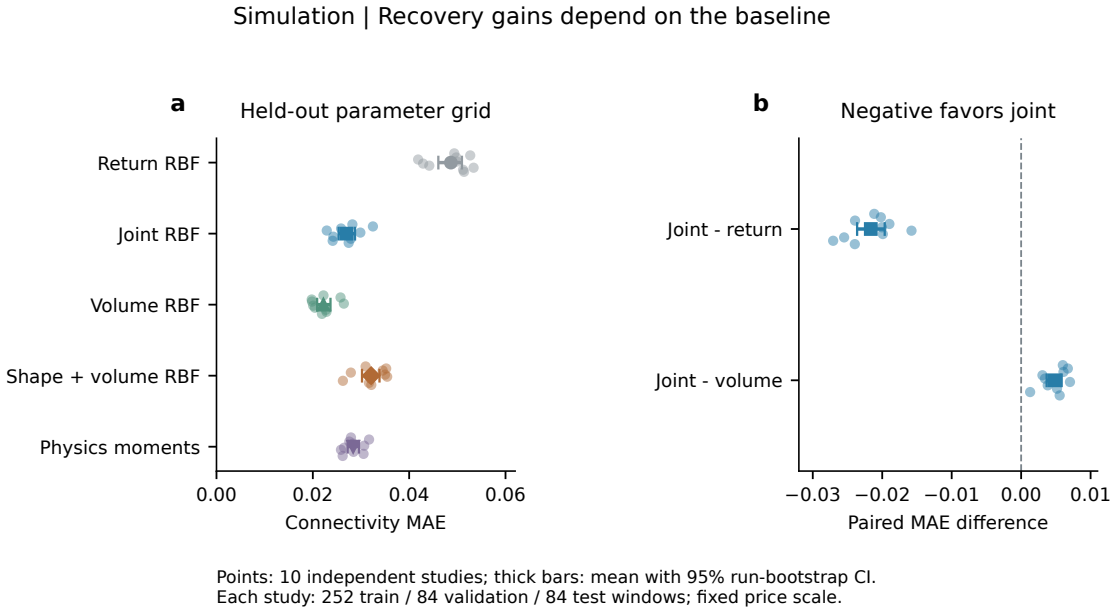


Figure 4. Parameter recovery across ten independently regenerated and retrained studies. Points are studies; summaries and intervals describe across-study variation. Table 2 gives grid and unseen-parameter MAE under the matched-scale design. [D01]

from 0.027013 to 0.258133. The scale-invariant representation and moment method remain at 0.032106 and 0.028372, respectively. Figure 5 shows this representation trade-off.

4.3 Even complete observations need not reveal internal edges

The chain $\{12, 23, 34\}$ and star $\{12, 13, 14\}$ differ as four-node graphs, yet both form a single cluster of size four. Equation (4) gives identical complete joint price–volume laws. More passive observations of this kind cannot recover the internal edges. This is stronger than failure of a particular classifier. Exact certificates for general cluster mixtures are also included with proofs and rational checks.

Parameter recovery, structural equivalence, and information lost through feature compression require separate treatment. Financial ABM calibration studies likewise distinguish reproducing stylized facts from identifying mechanism parameters[6, 7]. The characteristic function, known-truth recovery experiments, and market restrictions thus address structural information, estimation accuracy, and model compatibility, respectively.

Source tables and methods: `docs/theory/DEPTH_ADDENDUM.md`.

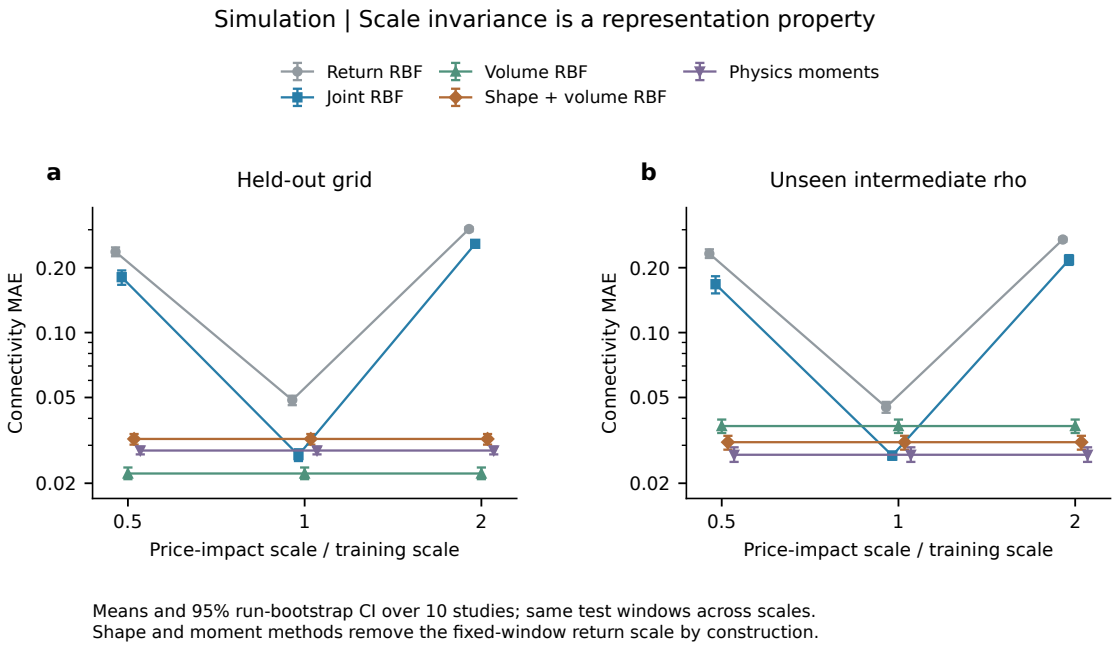


Figure 5. Price-response rescaling applied to the same evaluation windows in ten studies. Raw-return inputs change with scale, while moment inversion and scale-invariant inputs remove the specified window-level transformation by construction. Intervals resample studies. [D02]

5 Testing joint observations in Bitcoin

5.1 Data and price–volume comparison

The primary market case uses Binance Vision BTCUSDT spot five-minute bars[10]. The 2025 sample contains 105,120 bars and 365 complete UTC days, without filling or deletion. H1 provides calibration; H2 contains the original 184-day compatibility-test period. Returns are adjacent-close log changes, and bar volume is measured in BTC. Five-minute bars still compress the within-bar trade sequence. Figure 6 compares binned joint statistics and price-response proxies using total BTC trading volume.

Appendix Fig. 26 compares market and model tails on a calibration-defined scale. The market bars record price and volume, while trader counts, clusters and ρ remain unobserved. The empirical analysis therefore tests observable moment restrictions directly.

5.2 A joint necessary restriction derived from the mechanism

Eliminating scales from Eq. (3) gives a restriction for any graph distribution, beyond ER:

$$\Delta = \frac{1}{\mathbb{E}V}\left[\frac{\mathrm{Cov}(r^2, V)}{\mathrm{Var}(r)} - \frac{\kappa_3(V)}{\mathrm{Var}(V)}\right] = \frac{\mathbb{E}S_3}{N\mathbb{E}S_2} \in [1/N, 1]. \tag{10}$$

It requires fixed N, q, u, λ, symmetric independent actions, simplified execution, and linear response. Empirical calculations remove constant return drift and retain higher-moment and nonstationarity sensitivities.

For market H2, $\Delta = -8.181$, with an approximate 95% circular seven-day-block interval $[-11.998, -4.712]$. Figure 7 contrasts this estimate with the required positive range. The observed sign conflicts with the necessary range under the joint fixed-interface assumptions. All 12 calendar-month point estimates in 2025 are also negative, as are the checks excluding the highest-volume or highest-realized-variance day. These source-table sensitivities support using the restriction as a model diagnostic. The test identifies a failure of the fixed interface, while the next section examines how changing activity and response affects that restriction.

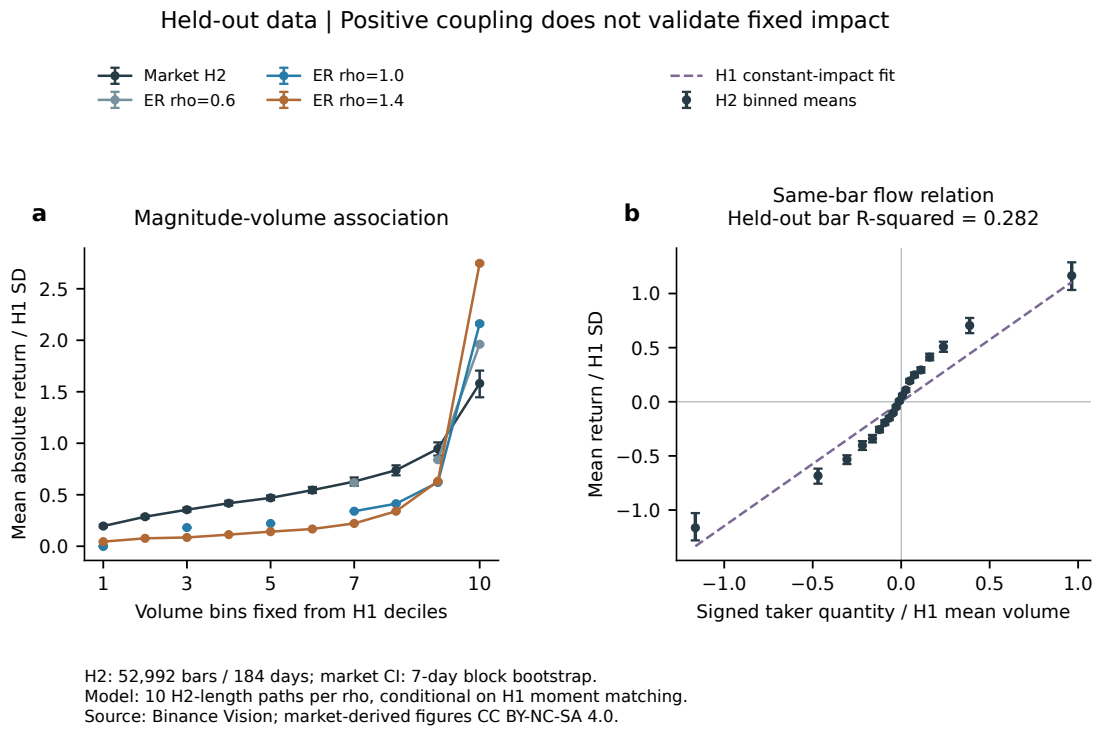


Figure 6. Bitcoin and model price–volume comparisons using H1-fixed bins. H2 magnitude–volume Spearman correlation is approximately 0.498, or 0.481 after seasonal adjustment. Empty model bins remain missing. Binance Vision, CC BY-NC-SA 4.0. [R01]

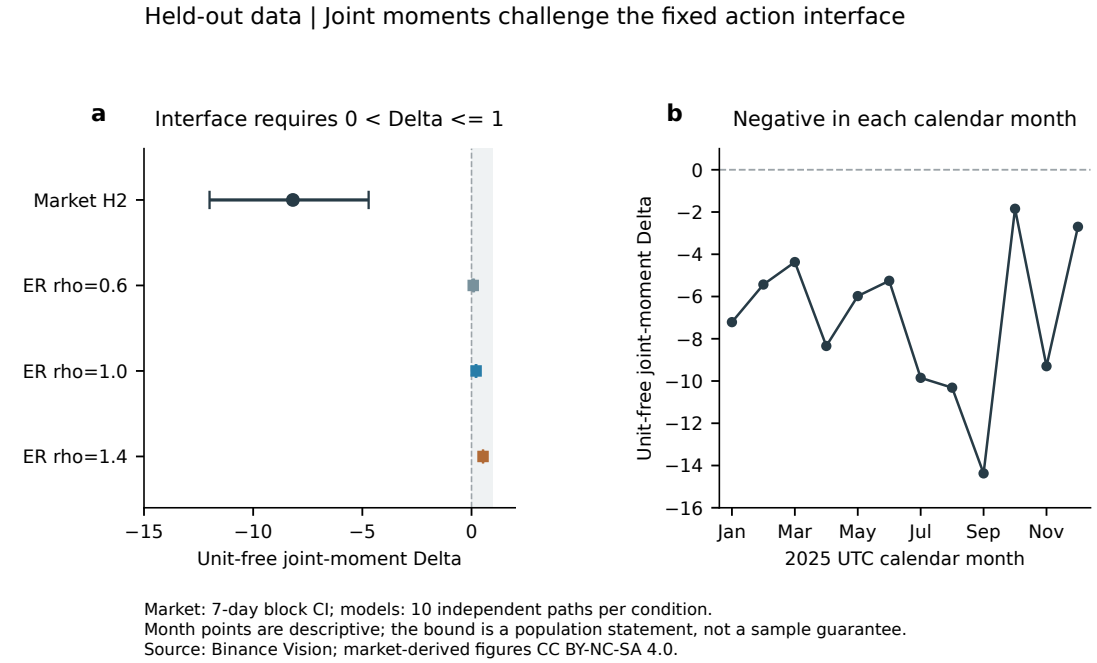


Figure 7. Joint-moment restriction for the fixed group-action interface and its BTC H2 estimate. The model requires a positive statistic. Market intervals use circular date-block resampling; monthly and extreme-day sensitivities are retained in the source tables. Binance Vision, CC BY-NC-SA 4.0. [R04]

5.3 Temporal structure supplies additional forecasting information

Let $\mathcal{F}_t$ denote the information available through period t. With fixed parameters and independently redrawn graphs and actions,

$$\mathbb{E}[r_{t+1}^2 \mid \mathcal{F}_t] = q(\lambda/N)^2 \mathbb{E}S_2 \tag{11}$$

is constant. This static baseline does not itself provide a date-varying next-period risk forecast. Market H2 has lag-one magnitude autocorrelation 0.344, signed-return autocorrelation about

−0.021, and a one-day aggregation variance ratio about 0.749. Temporal information therefore supplies an additional axis of mechanism comparison beyond the same-period cluster geometry.

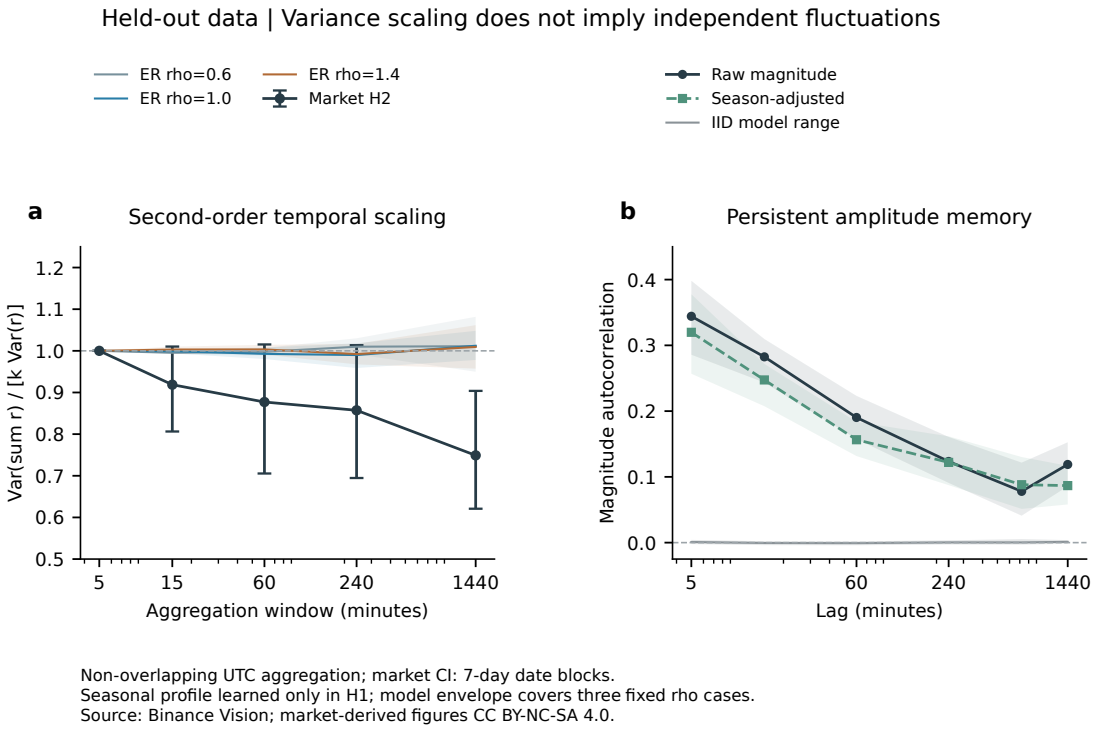


Figure 8. Temporal aggregation and lag dependence in independent model simulations and Bitcoin observations. Market magnitude memory contrasts with the independent-period baseline. Variance ratios and lag correlations describe temporal information, separately from cluster geometry. [R03]

6 Temporal states and forecasting information

To isolate temporal information, the controlled state extension keeps $N = 128, \rho = 1$ and independent graphs, while activity q_t or response λ_t follows an exogenous temporal state. Four arms each use ten paths of 8,192 periods, totaling 327,680 observations. Population mean volume and return second moments match across arms. Persistent activity produces magnitude and volume memory; persistent independent response can produce magnitude memory with nearly absent volume memory. Shuffling complete (r, V) pairs preserves the same-period empirical joint distribution while removing temporal memory. Figure 9 separates these informational roles.

Let $m_k = \mathbb{E}(q^k)$ and $A = \mathbb{E}S_2$, with activity independent of the graph. The total-variance identity gives

$$\mathrm{Var}(V) = u^2[(m_1 - m_2)A + N^2(m_2 - m_1^2)]. \tag{12}$$

The second term arises from changing activity, preventing interpretation of all volume variance as cluster structure. A complementary analytical construction, developed after the simulation comparison, sets $\lambda(q)^2 = \lambda_0^2 \bar{q}/q$, where $\bar{q} = m_1$ is mean activity. This keeps $\mathbb{E}[r_{t+1}^2 \mid \mathcal{F}_t] = (\lambda_0/N)^2 \bar{q} A$ constant. Volume may reveal activity states without improving this forecasting target. Its population $\Delta \approx -2.16348$ shows that the positive fixed-interface restriction does not extend to all herding mechanisms. This construction demonstrates how activity–response coupling can reverse the restriction; estimating such a mechanism in market data remains a separate inference task. Full assumptions and exact checks appear in STATE_MECHANISM_v2.md.

An observable can identify the current activity state while its usefulness for forecasting return variance depends on state persistence and response compensation. This distinction motivates evaluating temporal observables in the forecasting benchmark.

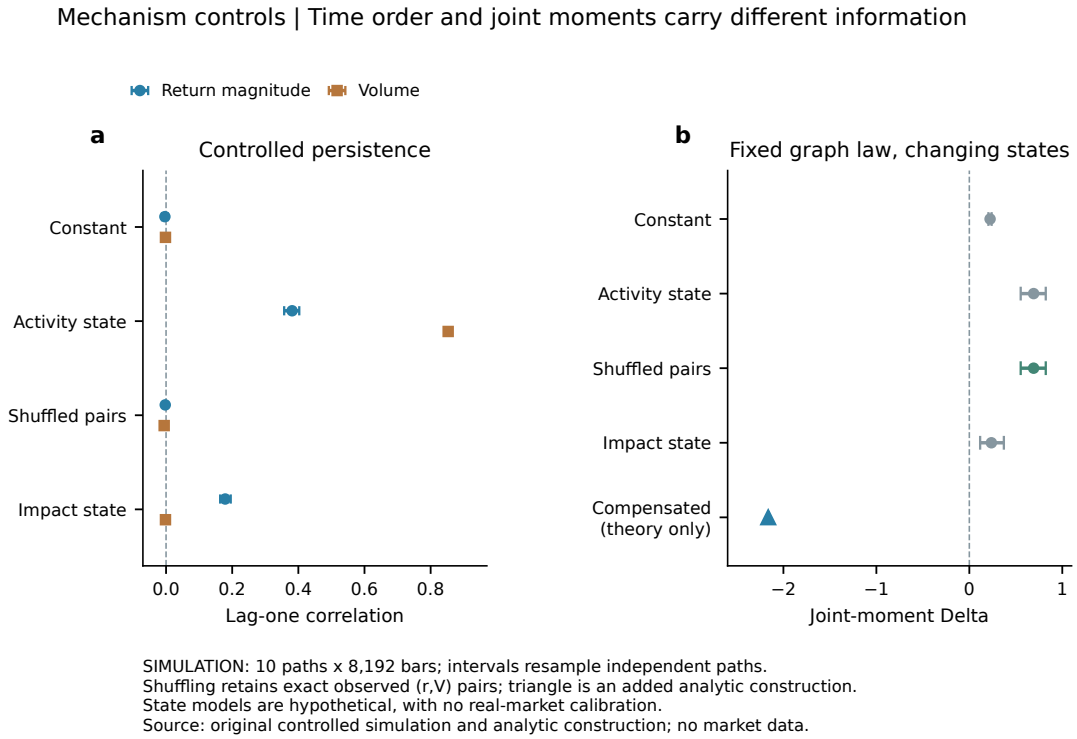


Figure 9. Controlled activity and response states under the same graph law. Persistent states produce different magnitude and volume memory; pair shuffling preserves the empirical same-period joint distribution and removes temporal order. Intervals resample independent paths. Sources: state_mechanism_v2. [V01]

7 Effective trading weights and the observable risk interface

The static group model supplies a tractable reference for a more general question: which statistical features of collective demand are needed for risk? Replace group size by a positive effective trading weight Q_C. A large decision, several investors acting together, and one parent order executed in pieces can contribute similar aggregate pressure. Their participant interpretations differ, so effective weights are defined in units of trading quantity rather than people.

7.1 The covariance mechanism survives the extension

For action vector ξ and fixed weight vector Q,

$$F = Q^{\mathsf{T}}\xi, \tag{13}$$

$$\begin{aligned}\mathrm{Var}(F \mid Q) &= Q^{\mathsf{T}}\Gamma Q \\ &= \sum_C Q_C^2 \,\mathrm{Var}(\xi_C \mid Q) + 2\sum_{C<D} Q_C Q_D \,\mathrm{Cov}(\xi_C, \xi_D \mid Q),\end{aligned} \tag{14}$$

where $\Gamma = \mathrm{Cov}(\xi \mid Q)$. The original independent symmetric-group law is the special case $\mathrm{Var}(F \mid Q) = q\sum_C Q_C^2$. With positive weights and marginal action variances fixed, additional positive cross-action covariance increases flow variance. Coupling can also change activity, mean direction, temporal dependence and price accommodation; its full price-risk effect is evaluated separately.

For execution sizes v_i in one bar, define the independent-sign reference

$$B = \sum_i v_i^2, \qquad G_V = \frac{\mathrm{Var}(F)}{B}, \qquad G_E = \frac{\mathbb{E}F^2}{B}. \tag{15}$$

The averaging window and execution unit must be declared. G_V is centered dispersion; G_E also contains mean-flow energy. Neither is a measured fraction of followers. Under the fixed symmetric sign reference, B is a size baseline; changing the grouping of executions changes that baseline.

A common-sign 100-unit parent split into ten 10-unit children has cumulative-flow variance 10,000 under a symmetric parent sign. Ten independent signs give 1,000. Both constructions have sum of squared child flows 1,000. In general,

$$\operatorname{Var}\left(\sum_t F_t\right) = \sum_t \operatorname{Var}(F_t) + 2\sum_{s<t} \operatorname{Cov}(F_s, F_t). \tag{16}$$

Thus directional persistence matters for cumulative risk, while it need not raise expected realized variance (RV) when per-bar second moments and a linear price response are fixed. This distinction determines the forecast target rather than serving as a cosmetic choice of risk statistic.

The execution bridge also yields a conditional inverse with heterogeneous individual sizes. Under independent symmetric group actions with common activity q, let i.i.d. positive individual marks have moments μ_1, μ_2, independently of the group structure and actions. Each active individual supplies one execution. For $\chi = \mathbb{E}S_2/N$,

$$G_E = 1 + \frac{\mu_1^2}{\mu_2}(\chi - 1), \qquad \chi = 1 + \frac{\mu_2}{\mu_1^2}(G_E - 1). \tag{17}$$

With equal marks, the amplification ratio equals group susceptibility. Exact finite enumeration and 300 independent simulation replications comprising 1.5 million periods verify the marked-size correction under its specified execution map. The map itself is identifying information: one participant splitting a two-unit parent into two unit fills and two participants sharing one common action can have the identical full (F, V, K, r) event law. The inverse concerns an effective group statistic under the execution assumptions, not a universally recoverable head count.

7.2 What the public execution data actually constrain

The measurement audit covers 663 complete Bitcoin days (190,944 five-minute bars) and 24 dates with microscopic labels. Raw and exchange-aggregate records reconcile with bar totals. Three earlier fixed-date cases contain 12,536,460 raw executions and 4,378,760 aggregate units. Exchange aggregation is an observed execution grouping, not complete parent-order reconstruction.

Nonnegative sizes imply

$$\frac{V^2}{K} \le B_{\text{raw}} \le V^2. \tag{18}$$

The mean ratio of these arithmetic bounds is about 22,754 on the complete daily records. Macro V, K alone therefore leave a broad size-concentration range. On sampled H2 dates, aggregate-execution B is on average 2.391 times raw-execution B. A January-only Gamma size calibration gives inferred energy amplification 45.85 versus observed 41.71 in July, but 56.39 versus 25.80 in October. The latter comparison exposes transfer error in the size baseline.

For a declared execution unit e with independent Gamma sizes of shape a_s, conditional normalized sizes are Dirichlet. Given its execution count $K^{(e)}$,

$$\mathbb{E}(B^{(e)} \mid V, K^{(e)}, a_s) = V^2 \frac{a_s + 1}{K^{(e)} a_s + 1}. \tag{19}$$

The working state rule targets exchange-aggregate B, predicting aggregate counts from the observed raw count through an H1-calibrated count map and using state-conditioned concentration. The fitted Gamma shape is a prediction parameter, not evidence that empirical sizes are i.i.d. Gamma. Relative to a refitted pooled-Gamma control, StateGamma reduces aggregate-B loss from .2691 to .2549 (5.28%). The 24-date audit reconciles approximately 88.1 million raw executions and 29.2 million aggregate units from 48 checksum-verified archives.

Across 12 held H2 microscopic-label dates spanning six months, mean daily absolute relative amplification error is 35.14%; providing observed aggregate counts reduces it to 18.18%, at higher information cost. Measured and inferred G_E share a numerator, so these discrepancies evaluate the execution baseline, not independent flow-variance recovery. Centered and uncentered statistics are reported separately.

Large count dispersion also has alternatives to coordination. In 256 independent repetitions per null construction, heterogeneous-intensity Poisson processes without self-excitation produce high Fano ratios. Their representative values, about 2,802 and 7,091 for varying-intensity and mixed-state constructions, coexist with a real count ratio near 13,784. A near-critical value obtained by substituting count moments cannot be identified with a market branching ratio or follower fraction.

Evidence record: measurement audit W1 and the fixed-date execution and macro-estimator amendments.

7.3 The risk map and its predictive requirement

For the original equal-size interface, $Q_C = us_C$ gives $r = [\lambda/(Nu)]F$, so the linear response below recovers its price law. For information $\mathcal{I}_t$, write $m_F = \mathbb{E}(F \mid \mathcal{I}_t)$ and a local response $r = \alpha + \beta F + \eta$. If α, β are fixed conditional on $\mathcal{I}_t$, $\mathbb{E}(\eta \mid \mathcal{I}_t) = 0$, and $\mathrm{Cov}(F, \eta \mid \mathcal{I}_t) = 0$, then

$$h = \mathbb{E}(r^2 \mid \mathcal{I}_t) = (\alpha + \beta m_F)^2 + \beta^2 \overline{B} G_V + \sigma_\eta^2. \tag{20}$$

This connects the group covariance mechanism to a return second moment. A same-window product $\overline{B}G_V$ is exactly the flow variance used to define it. Forecast information must therefore come from future state evolution, a cheaper valid measurement route, or a tested restriction that improves an information-matched forecast. Re-labelling a measured variance does not establish that increment.

8 Interdependent decisions, perturbations and future distributions

8.1 A weighted fast–slow decision network

The dynamic extension retains 48 effective units with fixed positive normalized weights w_i, two fixed sparse graphs and binary fast/slow states s_i^f, s_i^s. The fast layer responds to current neighbor pressure; the slow layer supplies a more persistent decision state. Each unit also carries an adaptation trace z_i. Interconnected opinion processes motivate this layered construction[13]; its financial interpretation and numerical rules are stated and tested here.

With capital-weighted neighbor direction u_i^h, the fields are

$$\begin{aligned} H_i^f &= J_f u_i^f + \kappa s_i^s + \rho_f \mu s_i^f - a_f \mu z_i + h_t, \\ H_i^s &= J_s u_i^s + 0.55\kappa s_i^f + \rho_s \mu s_i^s - a_s \mu z_i + 0.25 h_{\mathrm{exec},t}. \end{aligned} \tag{21}$$

The next fast sign is sampled with probability $[1 + \tanh(1.5H_i^f)]/2$ of being positive. The slow state attempts an update with probability 0.14 per step, and

$$z_i' = (1 - \epsilon) z_i + \epsilon s_i^{f,\mathrm{new}}, \qquad \epsilon = 0.055. \tag{22}$$

Here J_f, J_s are within-layer influence strengths, κ is cross-layer coupling, ρ_f, ρ_s are inertia coefficients and a_f, a_s are adaptation coefficients; these layer coefficients are distinct from the ER degree ρ. The inertia term favors retaining a direction; the adaptation term can reduce the attractiveness of a persistent orientation. Opposing positions are signs within this model. Layer coupling is same-unit fast/slow influence, not a measured institution-to-institution dependency or a rewiring rule.

Independent activity draws A_i with probability $a = 0.58$ generate background quantities $C_0 w_i A_i$, with $C_0 = 1200$. For an additional deterministic pulse d,

$$F = d + C_0 \sum_i w_i A_i s_i^{f,\mathrm{new}}, \qquad V = |d| + C_0 \sum_i w_i A_i. \tag{23}$$

The model count $K = \sum_i A_i$ records active effective units. It is not the number of exchange executions. A positive accommodation state L and source noise convert flow into return through the frozen source law,

$$r = \mathrm{clip}\left(0.009 \tanh\left[\frac{0.34(F/1200)}{L} \right] + \eta, -0.018, 0.018 \right). \tag{24}$$

Accommodation is a simulated response state, not an observed order book. This law allows both flow alignment and absorption to affect price fluctuations. It does not contain an estimated intrinsic-value process.

8.2 A perturbation can induce trading without increasing every risk target

Paired-seed interventions compare the same background innovations with and without a pulse. Across 128 paired seed units, a 24-unit buy pulse raises first-step flow by 33.817 on average, with 95% paired interval $[30.062, 37.987]$. Subtracting the direct pulse leaves an induced background response of about 9.817 units. Thus the model gives a concrete route from a large trading decision to additional aligned trading.

The 48-step RV difference has mean -1.50×10^{-7} and interval $[-4.94\times10^{-7}, 1.36\times10^{-7}]$. At eight fixed latent validation states, the same positive pulse raises next-step flow second moment at four states and lowers it at four: a buy pulse can partly offset an initially negative state. These results distinguish a local response from a universal volatility increase. Without a scheduled opposite parent program, 108 of 128 seeds cross between opposite alignment thresholds. Such random internal switching is a model outcome, while strategic early exit and an inevitable consensus-to-crash cycle remain separate hypotheses.

8.3 Exact one-step moments and distribution-preserving propagation

Let $t_i = \tanh[1.5H_i(S, d)]$ for complete state S, with the common-input channel off. Independent transition and activity draws give

$$m_F(S, d) = d + C_0 a \sum_i w_i t_i, \tag{25}$$

$$v_F(S, d) = C_0^2 \sum_i w_i^2 (a - a^2 t_i^2). \tag{26}$$

When a common future input is present, these formulas are conditional on that input, which must subsequently be integrated. Exact conditional F, V, K means and covariances provide the observation interface. One-step Monte Carlo agrees with the formulas, and a 144-dimensional conditional-mean Jacobian agrees with finite differences to a maximum error of about 1.62×10^{-10}.

Repeated substitution of the mean state is a different operation: in a nonlinear model, $\mathbb{E}[f(S)]$ need not equal $f(\mathbb{E}S)$. A pair of hidden arrangements can have equal current F, V, K and aggregate fast/slow directions but different next-step mean flows, 412.47 and 566.37 in a retained example. Those current summaries are therefore insufficient for exact closure.

The operational solution propagates complete stochastic states and uses their joint future paths. A verification design uses eight new latent states and three pulse conditions, with 12,288 prediction paths, 98,304 independent reference paths and 196,608 source-engine one-step branches. For the unpulsed 48-step flow mean, RMSE normalized by background quantity is 0.00785 for stochastic propagation and 0.38842 for mean iteration (Fig. 10). These are numerical propagation errors against an independent simulation reference, not forecast losses on market data. The result supplies a tested multistep continuation of the original network law.

Evidence record: network perturbation W2; exact mathematical interface and propagation D01.

9 Shared strategy signals and feedback on the same network

Similar inputs across trading strategies give an economic source of correlated decisions. Rather than attaching a separate predictor, we add a common signal to the retained decision fields and compare it with independent node signals having identical marginal distributions. Positive and negative lagged price feedback, together with a common activity rule, are tested as distinct channels.

The 2007 quant episode provides an economic motivation: factor-portfolio and transaction evidence is consistent with overlapping deleveraging and temporary withdrawal of market-making capital[16]. Funding/market-liquidity theory also supplies a conditional feedback channel between the ability to finance positions and the ability to absorb trades[17]. These precedents motivate separate shared-input and response mechanisms. Our experiment does not infer actual institution identities, reconstruct their portfolios or implement their complete margin rules.

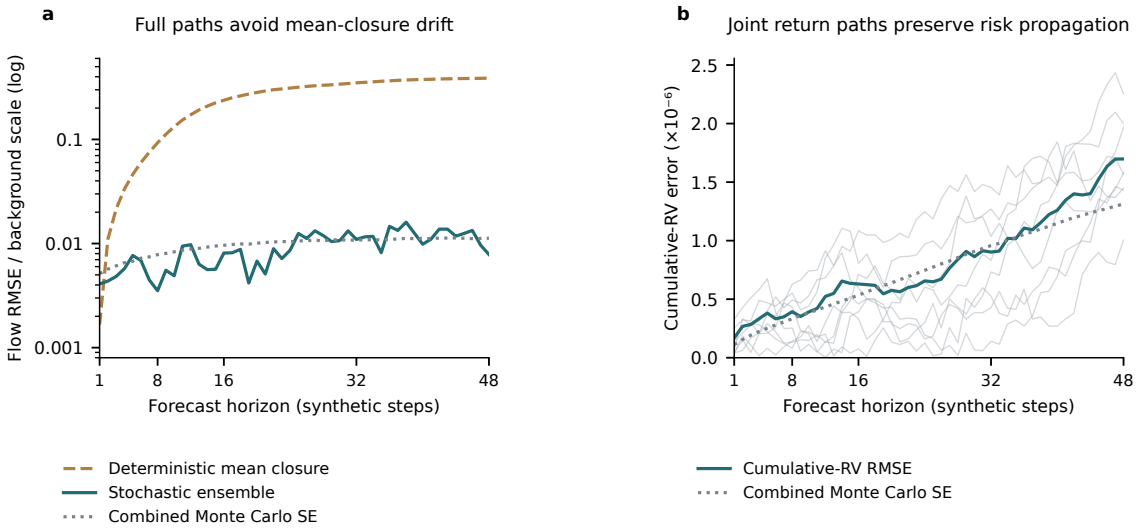


Figure 10. Full-state stochastic propagation and mean-state iteration against independent large-ensemble references. Eight fixed validation states, three pulse conditions and all declared horizons are retained. Monte Carlo agreement evaluates the propagation calculation, separately from latent-state inference and market forecast accuracy. Source: controlled continuation, D01.

9.1 A conditional link from shared information to covariance

For a shared standard Gaussian Z, positive loading $g = 0.15$ and the same fixed prior state, define $t_i(Z) = \tanh[1.5(H_i + gZ)]$. The independent-signal alternative substitutes independent Z_i. Node marginals and mean flow match, while

$$\begin{aligned} &\mathrm{Var}(F \mid S, \text{shared}) - \mathrm{Var}(F \mid S, \text{independent}) \\ &\quad = 2C_0^2 a^2 \sum_{i<j} w_i w_j \,\mathrm{Cov}_Z(t_i(Z), t_j(Z)) \geq 0. \end{aligned} \tag{27}$$

Both functions increase in the same scalar input. With an independent copy Z', their covariance is one-half the expectation of $(t_i(Z) - t_i(Z'))(t_j(Z) - t_j(Z'))$, a nonnegative product. This standard covariance argument connects a shared strategy input directly to Eq. (14). Opposite loadings and changing activity or accommodation are outside this conditional inequality.

Numerical quadrature at all eight frozen validation states gives a one-step flow-variance increment of 24.1–35.5% relative to independent signals; 64- versus 128-node quadrature discrepancies are below 1.82×10^{-11} in absolute variance units. Multiperiod return risk is then evaluated through the actual stochastic network rather than inferred from the one-step sign alone.

9.2 Paired economic-channel experiments

Six arms, including the baseline, generate 768 complete 240-step paths from 128 paired seed units; a further 64 baseline paths supply independent reference training. Table 3 reports all five interventions, with the ten RV/alignment comparisons retained under their frozen multiplicity rule.

Table 3. Controlled economic channels in one fixed network configuration. RV changes are percentages of baseline mean RV; alignment changes are percentage points.

Channel	RV change	Alignment change
Common signal	+3.62%	+3.92
Independent signals, same marginals	−1.78%	−0.52
Positive lagged price feedback	+6.24%	+4.31
Negative price feedback/accommodation	−3.99%	−2.74
Common activity contraction	−6.20%	0 exactly

Independent-signal uncertainty intervals cross zero in the primary changes. The common activity rule reduces participation without altering the direction equation, so its alignment statistic is exactly unchanged. It is not a forced-liquidation or strategic-exit mechanism. The experiment identifies how specified shared inputs and feedback alter risk inside this network configuration; the corresponding causal attribution to real quant institutions would require additional evidence.

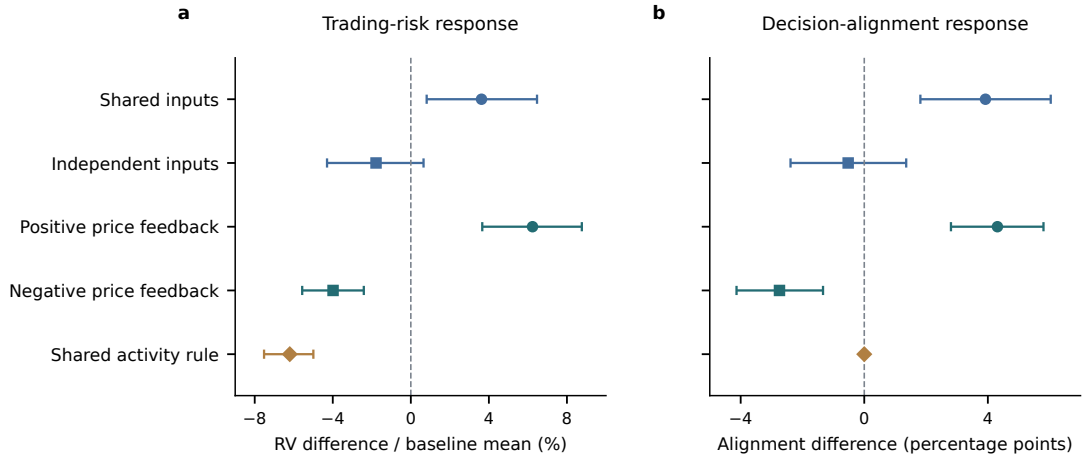


Figure 11. Paired shared-input, independent-input, feedback and activity interventions. The same seed supplies background innovations across arms. Both primary outcomes and every declared intervention are shown; the results concern this fixed source-law configuration.

Evidence record: economic-channel design N01, all paired contrasts and common-input quadrature identity.

10 Observable-state inference and coherent future risk

10.1 From an observable prefix to a distribution of effective states

The inverse task now asks for risk-relevant effective state rather than every edge or participant. A student estimator receives past F, V, K, r; true weights, graphs, node states and future paths are withheld. The supplied 48-unit law and graph/weight distribution are explicit prior information. The first causal particle implementation uses 64 training, 32 selection and 64 new test trajectories. Future-data mutation leaves its predictions unchanged. Static-parameter resampling, however, reduces every tested graph/weight ancestry to one source, yielding zero-width HHI intervals. This implementation check motivates preserving multiple structural possibilities rather than interpreting a single remaining graph as identification.

A conditional simulation ensemble uses the same unchanged network law. Its bank contains 2,048 independent full trajectories, with separate 32-trajectory tuning, 64-trajectory calibration and 128-trajectory evaluation sets. Only 13 observable summaries of the past 32 bars are used to condition the bank. Standardization is fitted on the bank; neighborhood size is chosen among 32, 64 and 128 using tuning-set eight-step QLIKE, selecting 128. Every evaluated case retains 128 distinct source trajectories and their full future joint paths. This is approximate conditional simulation, not an exact Bayesian posterior.

For common nonnegative normalized scenario weights ω_j, the next-step flow moments use total variance:

$$\bar{m}_F = \sum_j \omega_j m_F(S_j), \tag{28}$$

$$\mathrm{Var}(F \mid y) \approx \sum_j \omega_j v_F(S_j) + \sum_j \omega_j [m_F(S_j) - \bar{m}_F]^2.$$

The second term carries uncertainty about hidden state. Table 4 evaluates that state estimate on known-truth simulated units, not on unobserved market participant labels.

Table 4. Conditional simulation state estimates on 128 independent evaluation trajectories. The unconditional comparison uses the same simulation bank. Coverage concerns raw 90% state intervals.

Target	Conditional RMSE	Prior RMSE	Coverage
Fast direction	.17567	.39313	95.31%
Slow direction	.17015	.42501	96.09%
Adaptation memory	.09030	.28313	98.44%
Accommodation	.12295	.15641	88.28%
Weight concentration (HHI)	.00477	.00505	83.59%

HHI intervals have mean width .00977, retaining uncertainty rather than a single repeated ancestor. Their 83.59% coverage is below nominal 90%; source diversity does not establish calibrated structural recovery. The simple F/V fast-direction proxy has RMSE .13519, so the next inferential target is incremental information in slow state, memory, concentration and accommodation. The two inference designs use different evaluation samples and are not a same-sample before/after improvement experiment.

10.2 Generate return and flow moments from common paths

Network RV forecasts average squared returns generated by the nonlinear source law. A local decomposition $r = \alpha + \beta F + \eta$ checks their accounting without assuming residual orthogonality:

$$\begin{aligned}\mathbb{E}r^2 = \alpha^2 + 2\alpha\beta\mathbb{E}F + \beta^2\mathbb{E}F^2 + \mathbb{E}\eta^2 \\ + 2\alpha\mathbb{E}\eta + 2\beta\mathbb{E}(F\eta).\end{aligned} \tag{29}$$

Common weights retain joint flow, accommodation, return and residual paths and preserve covariance positive semidefiniteness. All conditional-ensemble forecasts are positive without clipping. In contrast, fitting cross moments separately had produced 40 non-PSD cases in 1,192 selected rolling fits and 12 nonpositive calibrated risks. The coherent-path construction fixes that numerical inconsistency while leaving forecast accuracy to a separate test.

Expected H-step RV is $\sum_{h=1}^{H}\mathbb{E}(r_{t+h}^2 \mid \mathcal{I}_t)$. The second moment of a cumulative return additionally contains cross-time terms. These two outputs remain distinct throughout the experiments.

10.3 Separate predictive uncertainty from latent-state uncertainty

After freezing a positive point predictor $\widehat{h}(x)$, independent calibration units provide scores $s_i = |\log(y_i/\widehat{h}_i)|$. For nominal coverage $1-\alpha$,

$$\begin{aligned}k = \lceil (n+1)(1-\alpha) \rceil, \quad q_\alpha = s_{(k)}, \\ C(x) = [\widehat{h}(x)e^{-q_\alpha}, \widehat{h}(x)e^{q_\alpha}].\end{aligned} \tag{30}$$

For $k = n+1$ the order statistic is defined as infinity. With $n = 64$ and $\alpha = .1$, $k = 59$. This standard split-conformal construction[14] gives marginal coverage under exchangeability of independent calibration and evaluation units, conditional on earlier fitting/selection. It is not state-conditional coverage or a guarantee for dependent market dates. Risk-interval calibration does not calibrate HHI intervals.

A frozen 48-step particle-model evaluation uses 64 fresh calibration and 128 fresh evaluation trajectories. Raw intervals cover 118/128 in that new evaluation; calibration covers 121/128 with a larger mean width and slightly worse Winkler score. Its point QLIKE is .03594, compared with .03562 for the unconditioned source-law prior, with paired difference interval $[-.003365, .003888]$. Figure 12 retains this earlier particle-model test, separately from the conditional-ensemble comparison below.

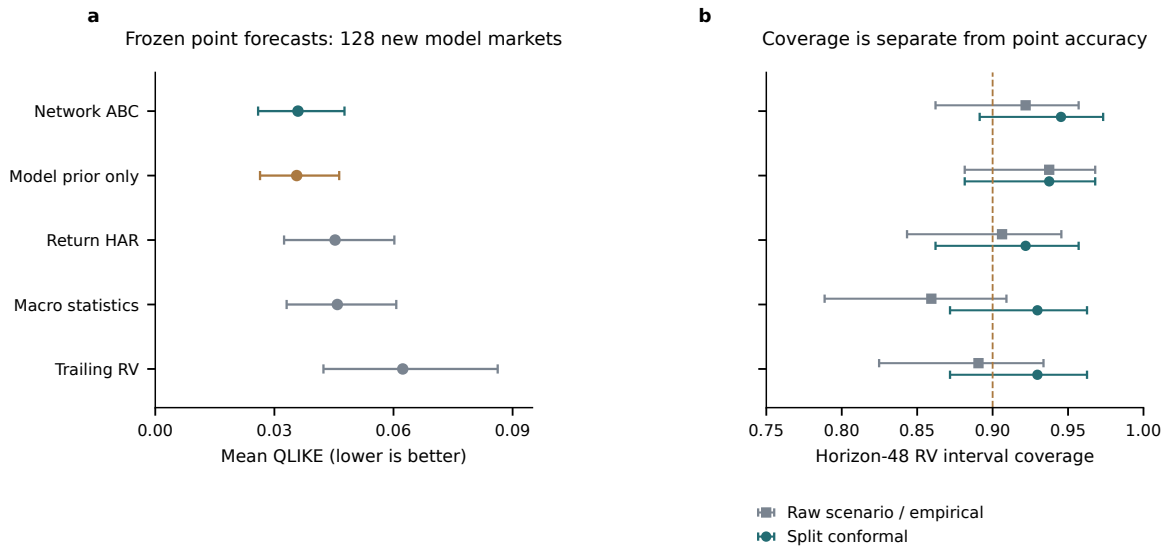


Figure 12. Observable-only inference and joint risk in the particle-model experiment. Source-law priors, test-state validation, interval width and calibration are displayed separately. This figure uses the first inference design and its frozen new calibration/evaluation sample; Table 5 uses a distinct conditional-ensemble design.

10.4 Risk comparison with equal simulation labels and macro features

The conditional ensemble is compared with a macro Gamma model using the identical 13 observable features and the same bank of future-risk labels. Fitted controls use the same bank and future-risk labels; the unconditional prior averages that bank, while trailing RV is a prefix-only baseline. Every method receives the same 64 interval-calibration labels. Eight steps is the fixed primary horizon, with one and 48 retained as secondary horizons.

Table 5. Primary eight-step risk on 128 independent controlled evaluation trajectories. QLIKE is smaller for better point forecasts. Coverage and width follow the shared 64-unit interval-calibration budget.

Method	QLIKE	Coverage	Mean width
Conditional network ensemble	.13412	95.31%	3.8×10^{-5}
Same-feature macro Gamma	.13329	94.53%	4.0×10^{-5}
Unconditional network prior	.13688	95.31%	4.1×10^{-5}
Return-HAR	.13570	95.31%	4.0×10^{-5}
Trailing RV	.19185	94.53%	4.9×10^{-5}

The paired QLIKE difference between the ensemble and Gamma is +.000837 (95% interval $[-.004832, .006098]$). Relative to the unconditional prior, the difference is −.002756 (interval $[-.015453, .008894]$). Incremental risk value over those two primary references is not established. Improvement over trailing RV survives the frozen four-comparison Holm rule, but that weaker reference does not replace the equal-feature and prior controls. At 48 steps, point scores .04304 and .04438 for ensemble and Gamma are reported as secondary outcomes, not a replacement primary horizon.

The full generative law, model age, unobserved initialization, offline state/risk labels, retained history and calibration are counted as information costs. A 32-bar deployed interface does not by itself establish savings over a statistical method using a longer observed history. The conditional-ensemble experiment establishes a working macro-to-state-to-risk calculation and a direct test of its added predictive value under disclosed simulation information.

Evidence record: observable inference I01, joint risk R01, calibration R02; conditional-ensemble state/risk scores and independent numerical recalculation.

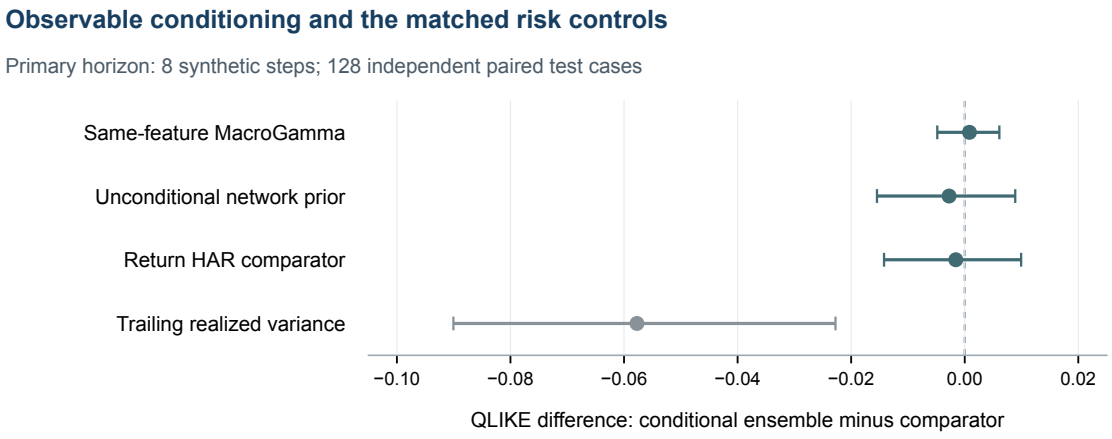


Figure 13. Primary eight-step conditional-ensemble risk contrasts: QLIKE difference from each prespecified reference, with 95% paired basic-bootstrap intervals; negative values favor the ensemble. The design uses 2,048 bank, 32 tuning, 64 calibration and 128 evaluation trajectories. The same-feature Gamma comparison shares observable summaries and simulated risk labels; the known generating law and offline labels remain disclosed costs. The three stronger-reference intervals cross zero. Trailing RV is a weaker comparator. Source: conditional-ensemble risk contrasts.

11 Connecting the network clock to real market observations

11.1 The earlier direct-map diagnostic

A decision unit, an execution and a calendar-time bar are different objects. A common-sign parent can generate multiple exchange executions; a five-minute bar can contain many internal decisions. Financial transaction models treat arrival times and price changes jointly[15], providing a precedent for a separate observation clock. Here the original network continues to supply directional interaction, while the arrival/fragmentation layer introduced below maps them to market observations.

A descriptive transport check compares all 663 previously examined Bitcoin days, partitioned into 1,989 blocks of 96 five-minute bars, with 64 new 96-step source-network blocks. Table 6 summarizes scale-independent discrepancies under the tentative direct step/bar map.

Table 6. Uncalibrated direct-map observation comparison. Means concern the stated block summaries; three blocks share each market date and are not independent experimental replications.

Block statistic	Source network	Bitcoin
Lag-one correlation of pressure F/V	.709	.142
Coefficient of variation of volume	.150	.928
Coefficient of variation of count	.123	.626

The direct map carries excessive directional persistence and too little activity variation. Raw exchange counts also lack a calibrated relation to 48 effective active units, and 31 real return bars exceed the unchanged source return cap. The model was therefore not evaluated as an independent new-market predictor. These diagnostics localize the observation problem rather than treating normalized features as a substitute for units, clocks and response calibration.

11.2 Calendar-time observation and price-response extensions

The direct-map result above concerns the unchanged source law in Eq. (24). Subsequent work has implemented the separate observation layer it motivated. The retained 48-unit fast/slow network evolves through eight decision events per five-minute bar; an activity and execution layer produces calendar-time F, V, K, and the price layer separates realized impact from recovery of predictable impact. The M2 extension additionally supplies correlated residual decision innovations

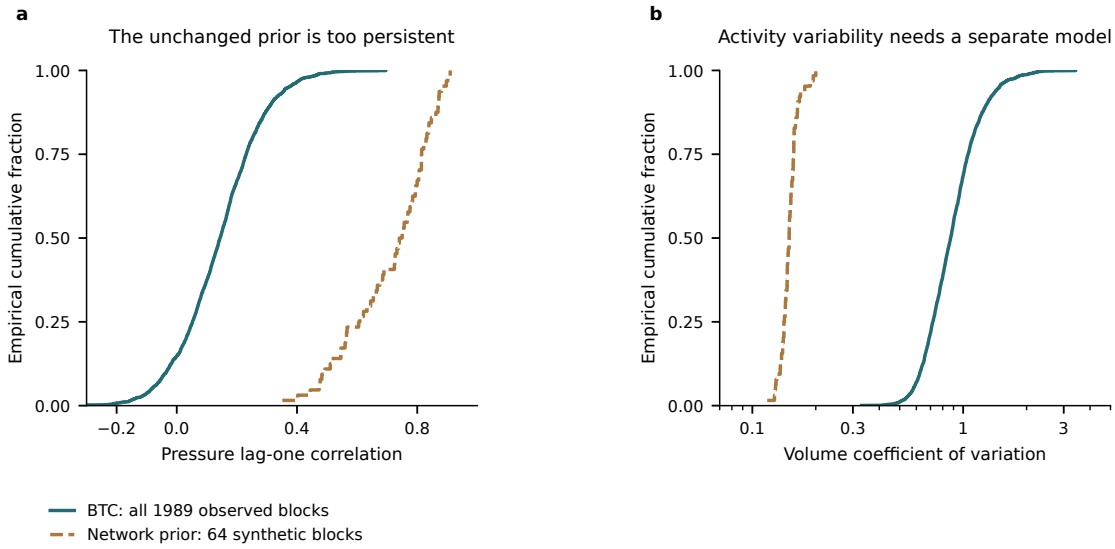


Figure 14. Source-network and Bitcoin observation checks under a tentative direct map. All 663 examined days and 64 new source-network blocks are used. Pressure persistence, activity variation, execution units and return support diagnose the observation layer before an independent market test is opened. Binance Vision-derived statistics retain source attribution.

and feedback from actual calendar returns. These changes preserve the network mechanism while explicitly changing its observation and response law. Results under that extension are therefore reported separately from the earlier direct-map comparison.

The price-scale repair uses the population identity $\mathbb{E}[g^2] = 1$ for volume-scaled innovations, rather than an accidental finite-path average. Under the resulting nominal calibration, five-, fifteen- and sixty-minute return-second-moment ratios to Bitcoin are 1.063, 1.144 and 1.273; 30 of 36 fixed descriptive conditions pass. The nominal five-minute ratio had been about 1.570 in the preceding calibration. The revised model still misses pressure persistence, volume variation and the sixty-minute second moment. The impact coefficient's relative Monte Carlo standard error is 8.32%, above the declared 3% precision threshold, and the finite parameter ensemble does not improve its aggregate mismatch score. Thus a usable, frozen calendar-time generator has been constructed, while precise market calibration remains incomplete. Sections 12–13 use its nominal parameters to investigate conditional risk and the information problem rather than treating that generator as an identified real-market law.

The practical criterion remains lower total information cost at comparable reliability, or better accuracy with the same observable fields and history. If an extra liquidity observation is proposed, its identification role and its availability to every comparator must be explicit. The following studies first locate the state information that could justify that addition and then evaluate a macro-only approximation against observed Bitcoin risk.

Evidence record: earlier direct-map comparison V01; population price-scale repair and M1/M2 calendar-time continuation; source tables retained with the study records.

12 Coordination risk depends on the receiving environment

The weighted covariance mechanism in Eq. (14) determines how collective actions change flow variation. Price risk also depends on the environment through which that flow is transmitted. We now test this link on the retained weighted network in calendar time, rather than interpreting more coordination as a context-free volatility multiplier.

12.1 A common calendar clock and a controlled intervention

Let ϱ denote correlation between future cross-agent residual decision innovations, κ_r the coefficient of actual calendar-return feedback into decisions, and L normalized latent shock capacity. These are

different from the original ER degree ρ and fast/slow layer coupling κ. A higher L represents greater model capacity to absorb a shock; it is not an observed dollar order-book depth. The predictable-impact subtraction coefficient χ concerns the price response, separately from return-to-decision feedback.

The inherited calendar price equation can be written

$$r_j = \alpha + \beta I_j + \sigma\sqrt{g_j^2}\,\epsilon_j, \qquad g_j^2 = V_j/\overline{V}_{\text{fit}}, \tag{31}$$

where I_j is realized impact minus predictable-impact recovery, formed before the current independent price innovation. The source records name this quantity Y_j; I_j keeps it separate here from a risk outcome. Actual noisy returns still enter subsequent decisions. The risk target is $Z_{t,h} = \sum_{j=1}^{h} r_{t+j}^2$, with $h = 12$ five-minute bars, and $\mathcal{R}_h(S) = \mathbb{E}[Z_{t,h} \mid S]$ for complete state S.

A future-covariance intervention preserves every unit's initial complete decision signal by remapping its idiosyncratic component, then changes only the covariance of subsequent residual innovations. Paired arms share activity, execution and pricing innovations. Their V, K paths and activity-scaled price-noise paths match exactly; all nuisance price parameters remain fixed. The comparison therefore asks how a change in future alignment propagates through the same initial market environment, rather than comparing different amounts of trading or refitting impact to each outcome.

12.2 Separating state heterogeneity from simulation noise

For each of 256 new independent M2 origins, three paired arms—full dynamics, future covariance off and return feedback off—generate 512 future branches. Origins are split 128/64/64 for fitting, development and evaluation. The first and second 256 branches provide independent labels and risk targets. We reduce current-bar label noise with

$$\mathbb{E}[r_j^2 \mid \mathcal{F}_j^{\text{pre-price}}] = (\alpha + \beta I_j)^2 + \sigma^2 g_j^2. \tag{32}$$

The tower property makes the sum of these conditional moments unbiased for expected future RV. Earlier price innovations still alter later states; this is termwise integration, not a claim that a global Rao–Blackwell variance-contraction theorem applies to the full feedback path.

The mean full-risk estimate is 3.6401×10^{-5}. Removing future covariance lowers it by 2.4830×10^{-6}, with paired-origin 95% interval $[2.2879, 2.6836] \times 10^{-6}$, or 6.82% of mean full risk. Removing return feedback lowers it by 1.5640×10^{-6}, interval $[1.2895, 1.8202] \times 10^{-6}$, or 4.30%. Integrated paired-label conditional variances are .299 and .283 times their direct squared-return counterparts; cross-half effect correlations rise to .911 and .974 (Fig. 15). Independent-replicate effect covariances are positive, so the measured state variation is not explained solely by Monte Carlo error.

These effects have a positive mean and real heterogeneity, but their marginal between-state variance is only about .28% and .50% of the full expected-risk variance. Those ratios are neither incremental R^2 nor causal shares of risk. More precise effect estimates still worsen evaluation QLIKE when jointly added to activity controls or simple-state controls, by 5.65% and 10.99%, respectively; both adjusted tests give Holm $p = 1$. The missing forecast value therefore cannot be attributed principally to too few future branches in the earlier study.

12.3 A conditional risk surface, rather than one coordination multiplier

A fresh response-surface study uses 32 origins and 128 paired future branches per cell. Its fixed grid combines $\varrho \in \{0, .35, .65\}$, $\kappa_r \in \{0, .15, .30\}$ and $L \in \{.5, 1, 1.5\}$ at $\chi = .5$, with two additional recovery settings at the nominal cell: 29 scenarios in total. Marginal cell intervals and paired intervention intervals answer different questions; Table 7 reports the latter.

The same future innovation alignment produces a larger price-risk increment when the simulated receiving environment has less capacity. The result gives an explicit mechanism for why a crowded common response may matter especially during impaired market accommodation. It does not infer how many real institutions share a factor or the fraction of Bitcoin risk caused by them. Factor overlap would have to produce sufficiently weighted, directionally and temporally aligned trade innovations; the factor's name or the number of firms alone does not supply that measurement.

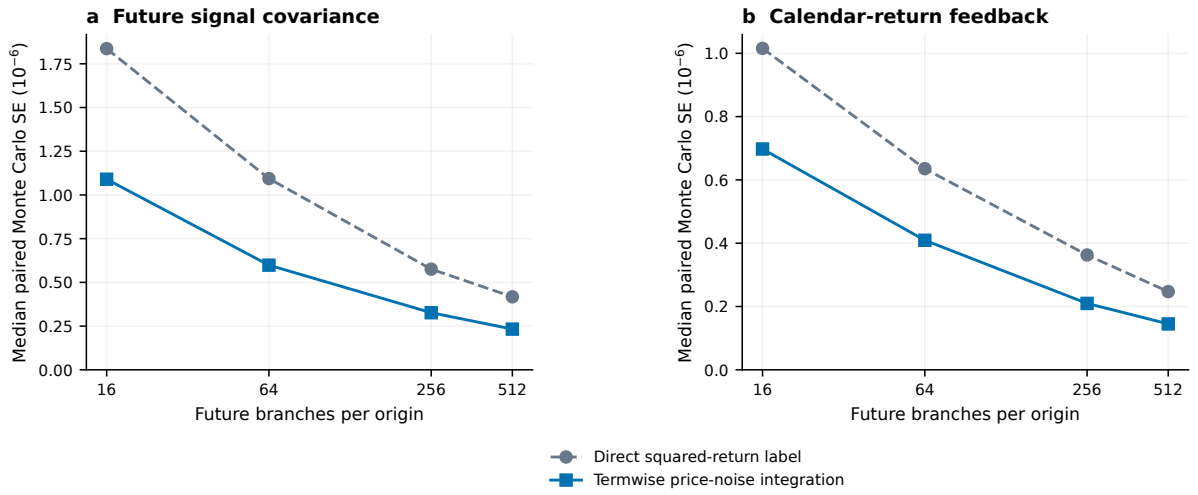


Figure 15. Independent-replicate precision diagnosis in the unchanged M2 dynamics. Each of 256 origins has 512 paired futures in each arm. Current-bar price-noise integration improves effect-label precision while preserving noisy future feedback. Average risk effects and cross-state heterogeneity are assessed separately from incremental forecast value. All quantities concern fixed nominal model parameters.

Table 7. Effect of increasing future innovation correlation from $\varrho = 0$ to .65 at $\kappa_r = .15$. Risk is the expected sum of twelve five-minute squared log returns. Intervals are paired-origin 95% basic-bootstrap intervals, with 4,999 replicates over 32 origins. Capacity is a latent model condition, not a measured market field.

Capacity L	Risk increment $\times 10^6$	95% interval $\times 10^6$	Increment/full risk
.5	3.3063	[2.7325, 3.8554]	7.44%
1.0	2.2303	[1.6799, 2.7258]	6.42%
1.5	1.6389	[1.1040, 2.0934]	5.42%

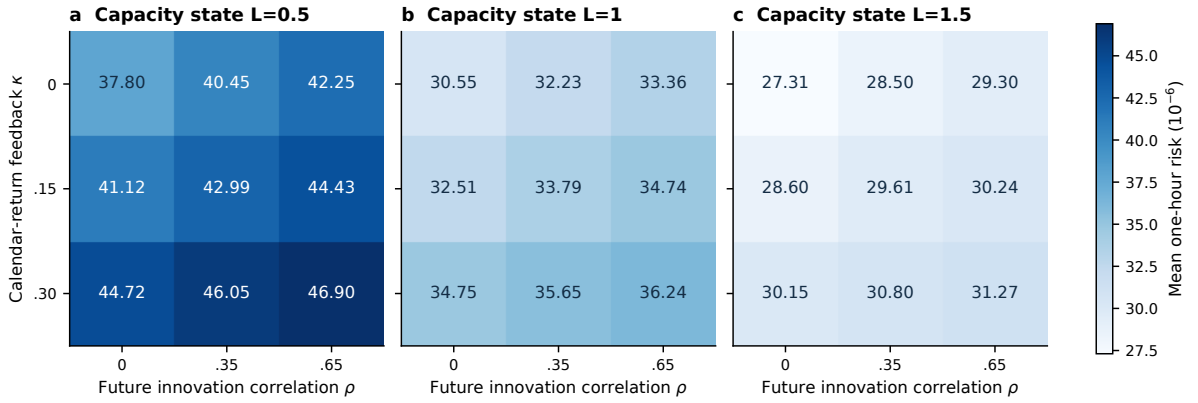


Figure 16. Expected-risk surfaces across fixed correlation, calendar-return feedback and capacity scenarios. The source data key `rho` corresponds to ϱ, the future innovation correlation, rather than the ER degree. Model states and interventions are distinct from observed Bitcoin quantities. All 27 main-grid cells and their source intervals are retained; two recovery cells complete the 29-scenario study.

The channels also interact. At $L = 1$, the covariance increment is 2.8116×10^{-6} with $\kappa_r = 0$ and 1.4836×10^{-6} with $\kappa_r = .30$. Their difference is -1.3280×10^{-6}, paired interval $[-1.8572, -.7096] \times 10^{-6}$. Feedback raises total risk while reducing the marginal covariance effect on this grid. This is a conditional subadditive interaction, not a general saturation law or evidence that stronger feedback is safe. Relative to $\chi = .5$, removing predictable-impact recovery raises risk by 2.8580×10^{-6}, interval $[1.4605, 4.0386] \times 10^{-6}$; increasing recovery to one does not establish an additional benefit. No optimal recovery rule follows from these three settings.

For a fixed observed prefix, changing the future environment produces a median maximum/minimum expected-risk ratio of 1.694 across tested contexts. This is a useful stress-scenario

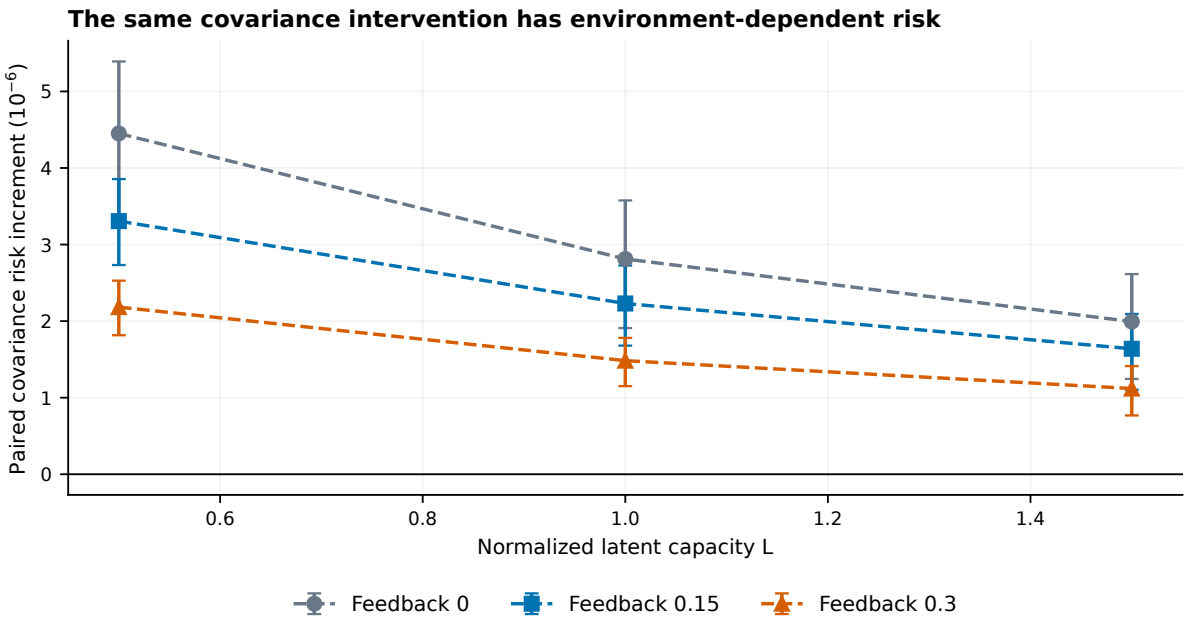


Figure 17. Paired covariance effects and environmental interactions. Error intervals resample whole simulation origins, preserving paired futures and shared paths. Lower shock capacity magnifies the tested covariance effect, while correlation and return feedback are not simply additive. Capacity, feedback and recovery are experimental model controls, not inferred policy thresholds.

range: it changes the future law and does not prove non-identifiability under one fixed stationary law. The explanatory object is consequently a state-and-environment risk map, not a unique coordination count recovered from one price–volume slope.

13 From mechanism effects to observable predictive states

13.1 Why a positive average effect is not a forecasting gain

Let X be the allowed past macro history, S the complete source state and Z future realized risk. The paired contrast

$$\Delta(S) = \mathbb{E}[Z^{\text{common}} - Z^{\text{independent}} \mid S] \tag{33}$$

compares two intervention worlds. Prediction within one world instead asks which observations discriminate its higher- and lower-risk states. A nearly constant positive Δ changes the level absorbed by that world's baseline forecast. A variable Δ can be redundant with activity or remain unrecoverable from the permitted observations.

This distinction is exact for the conditional-mean target of QLIKE. For positive risk Z, forecast a and $\ell(Z, a) = Z/a - \log(Z/a) - 1$, write $m_X = \mathbb{E}[Z \mid X]$ and $m_{X,W} = \mathbb{E}[Z \mid X, W]$ for additional state information W. Under finite expectations,

$$\mathbb{E}[\ell(Z, m_X) - \ell(Z, m_{X,W})] = \mathbb{E}[\log m_X - \log m_{X,W}] \geq 0. \tag{34}$$

Conditional averaging gives $\mathbb{E}[m_{X,W}/m_X] = 1$; conditional Jensen gives the inequality. The gain depends on within-world information about conditional risk, not the mean intervention contrast. Estimated learners need not attain this inequality, because finite-sample fitting and representation choices add error.

13.2 Which hidden state block adds information?

The independent-replicate design permits a direct information diagnosis. Each readout uses the same 128/64/64 origin split, a fixed Gamma/RBF learner menu and three penalty choices. Training labels use one set of 256 futures; evaluation targets use the independent second set. The target is average simulated future risk, which is less noisy than a single realized market outcome, so these losses are not directly comparable with the Bitcoin scores below.

Table 8. Effective-state information diagnosis at 64 evaluation origins. The hidden environment comprises day/intraday activity, size, liquidity and weight concentration. R^2 uses squared error relative to the fitting-period constant forecast, not the evaluation-period mean.

Information supplied to readout	QLIKE	R^2
Activity and past-volatility summaries	.028130	.93171
Complete 32-bar macro representation	.036879	.87441
Macro representation plus hidden environment	.008413	.95498
Above plus direction, memory and residual summaries	.008519	.96631

The environment block improves QLIKE over the full macro readout by .028467, with signed difference interval $[-.038889, -.016976]$ and Holm $p = .0006$. Its joint gain cannot be attributed to activity alone: size, liquidity and weight concentration are included. Adding direction, memory and residual summaries establishes no further QLIKE improvement (Holm $p = 1$). The higher last-row R^2 measures a different error and does not reverse that result. Activity summaries nevertheless estimate covariance and feedback effect variation with evaluation R^2 about .749 and .863. Recovering an effect is possible without that effect adding information beyond a strong risk baseline.

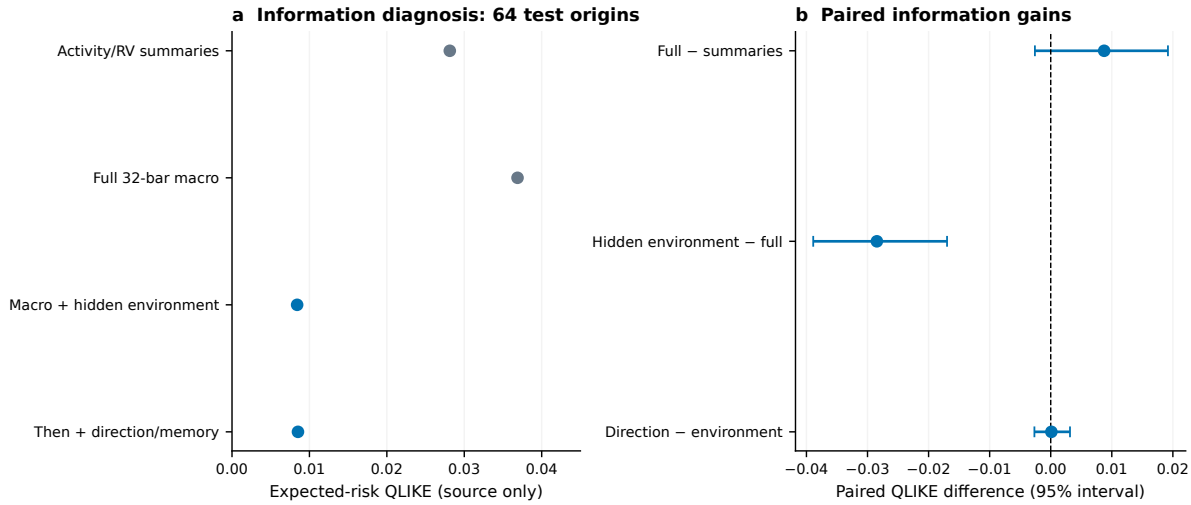


Figure 18. Risk information across matched source-origin and learner budgets. The environment block includes activity, size, liquidity and concentration; it is not an activity-only or observed-order-book intervention. Inferior performance of the full macro readout reflects finite-sample estimation, not a claim that supplying more raw information destroys information. Finite-neighborhood diagnostics are not treated as an identification impossibility theorem.

These results redirect the inverse task toward a joint effective state: pressure innovations, price response and environment-dependent residual risk. They do not support increasing simulation effort or adding another coordination scalar without a distinct information role.

13.3 An observable innovation–response closure

The network motivates a past-only macro approximation that separates persistence from new pressure:

$$p_t = F_t/V_t = \widehat{p}_{t|t-1} + u_t, \tag{35}$$

$$r_t \simeq \alpha + \lambda_o[p_t - \chi\widehat{p}_{t|t-1}] + e_t. \tag{36}$$

A pressure autoregression uses candidate lags 1, 4 or 12, previous actual return and clock terms; a finite fitting-period menu selects $\chi \in \{0, .5, 1\}$. A clock-aware log-volume/count model propagates activity. Local twenty-bar uncentered innovation second moments, with a zero conditional-mean approximation, supply a positive-semidefinite noise representation. First and second moments are propagated over the same twelve future bars through a linear state covariance recursion

$\Sigma_{j+1} = A\Sigma_j A^\mathsf{T} + Q_j$, including direct first-bar innovation. This is an effective moment closure, not exact latent-state recovery in the nonlinear network.

Persistent order flow and compensating price/liquidity response motivate this separation[18, 19]. Those studies concern their own market observations and do not validate this particular five-minute Bitcoin approximation. Here the closure inherits the network's pressure, feedback and impact-recovery interpretation; its forecast value is judged by the same observed target as the statistical controls.

The fitting period contains 181 dates and 3,801 risk origins, plus 47,060 deduplicated past-bar rows for self-supervised one-bar fitting. The first 145 dates fit the inner model and the last 36 select its finite menu before refitting the full period. The first 46 dates of the following development quarter select a risk readout; the last 46 calibrate intervals. The selected pressure lag is 12 and selected recovery is $\chi = 1$. This menu choice does not identify a true Bitcoin recovery coefficient. All methods receive the same F, V, K, r fields, thirty-two-bar history and fitting data; the mechanism adds derived representations, priors and computation, which remain part of its total cost.

13.4 Internal Bitcoin results and a representation control

At UTC 12:00 each day, the target is the next twelve five-minute squared returns. All 674 examined dates have appeared earlier in the project; the 401 scored dates are an internal reused evaluation, not new external confirmation. Seven fixed mechanism readouts and three Gamma penalties are developed without joining evaluation truth. Among RiskOnly, HAR+Risk and HAR+Components, development selects HAR+Components. Table 9 keeps that primary identity distinct from two controls added after the mathematical cancellation below was discovered.

Table 9. One-hour Bitcoin risk evaluation on 401 previously seen dates. Lower QLIKE is better. The first four rows belong to the fixed comparison; the final two are post-run descriptive representation controls and do not reset the primary test family.

Method	QLIKE	Role
HAR-All	.532309	Same-field reference
RawHistoryGamma	.539515	Developed empirical pipeline
HAR+Risk	.524478	Single-risk readout
HAR+Components	.501501	Selected primary candidate
HAR+RV20	.502745	Post-run control
HAR+RawMoments20	.499219	Post-run control

The selected candidate improves the QLIKE point value over HAR-All by 5.79%, but its paired loss difference is $-.030808$, with seven-day-block 95% interval $[-.078475, +.017982]$ and Holm $p = .33$. Fourteen- and twenty-eight-day blocks also fail to establish improvement. Six fixed primary questions use within-family Holm correction; paired circular bootstrap resamples whole dates in 4,999 replicates rather than treating overlapping windows as independent. The improvement over the developed raw-history pipeline is likewise uncertain (7.05% point value; Holm $p = .212$). The selected candidate and HAR-All both cover 94.51% at nominal 90%, with widths 1.1905×10^{-4} and 1.1831×10^{-4} and CRPS 1.2006×10^{-5} and 1.2301×10^{-5}. These metrics do not establish overall distributional superiority.

One frozen ablation supports predictable-impact subtraction within the *single-risk* readout: HAR+Risk improves over HAR+NoRecovery by about .52%, with Holm $p = .0156, .0264, .024$ for 7-, 14- and 28-day blocks. This is not a causal decomposition of the selected HAR+Components result or an observation of actual market coordination. Feedback removal establishes no gain; the residual-channel comparison is not robust to the longer blocks. Full scores and contrasts remain in the source tables.

13.5 An exact cancellation that limits the forecast interpretation

An acceptance check identifies an informative limiting case. With $\chi = 1$, constant local innovation moments and *activity prediction switched off*, set $e_t = r_t - \alpha - \lambda_o u_t$, $v = \mathrm{mean}_{20}(u^2)$, $\eta_o =$

$\mathrm{mean}_{20}(ue)/v$ and $z_t = e_t - \eta_o u_t$. Orthogonalization gives $\mathrm{mean}_{20}(uz) = 0$, so

$$(\lambda_o + \eta_o)^2 v + \mathrm{mean}_{20}(z^2) = \mathrm{mean}_{20}[(r-\alpha)^2]. \tag{37}$$

Complete recovery also removes predictable pressure from the future mean return in this closure. Its local-noise-only risk therefore becomes

$$\widehat{\mathcal{R}}_{12}^{\mathrm{local}} = 12\left\{\alpha^2 + \mathrm{mean}_{20}[(r-\alpha)^2]\right\}. \tag{38}$$

The identity is independently reconstructed on all 3,801 fitting and 674 query histories; maximum relative query discrepancy is 3.25×10^{-15}. It applies to this no-activity-forecast limit, not automatically to the full activity-scaled candidate.

This identity motivates the two explicitly post-run controls in Table 9. Adding ordinary RV20 nearly reproduces the main point improvement, and adding twenty-bar direct macro moments yields a lower point loss. They use fixed fitting/development procedures and do not become a new primary advantage test. The current improvement can consequently be explained by additional summaries or representations; it cannot be assigned uniquely to inferred coordination or network structure. The positive simulation mechanism, the useful conditional risk surface and this forecast boundary are compatible conclusions about different questions.

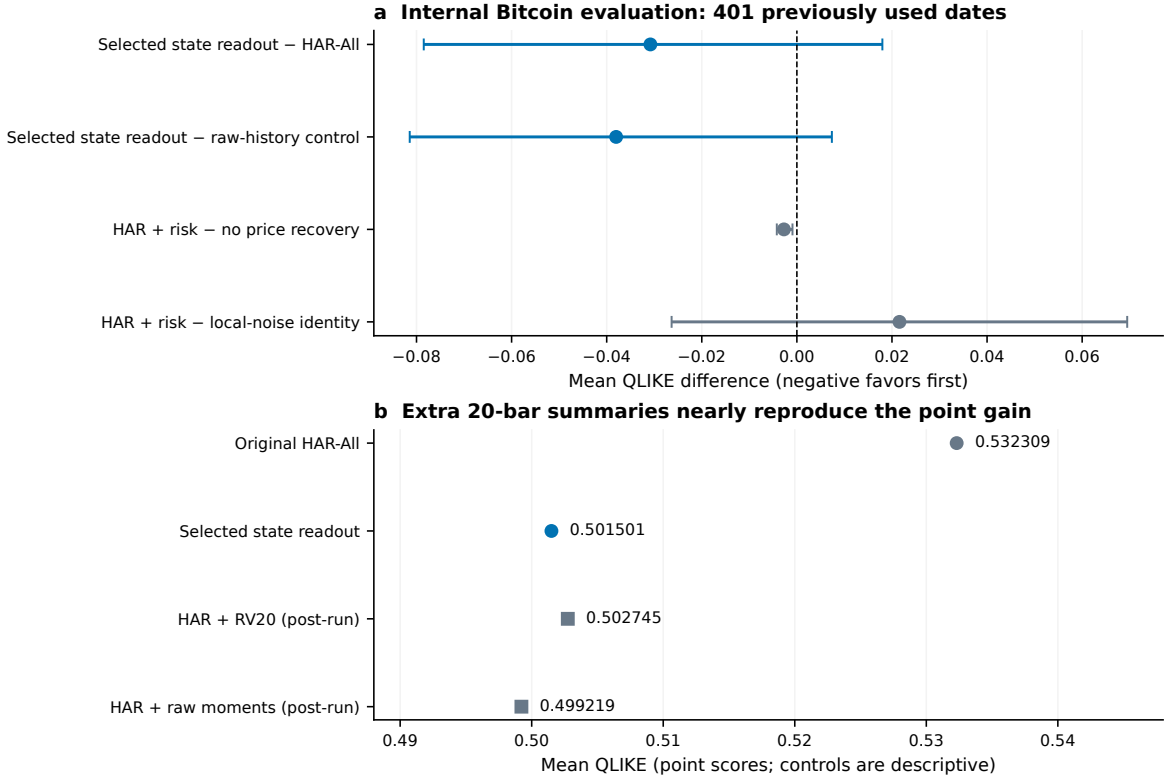


Figure 19. Observed internal Bitcoin forecast comparison and the closure's cancellation check. The selected candidate is distinguished from post-run RV20 and direct-moment controls. Evaluation dates are reused project observations; simulated environment cells are not superimposed as measured market states. The cancellation concerns $\chi = 1$ with constant local noise and activity prediction off.

Source tables and methods: `studies/state_environment_v9/`.

14 Historical next-day statistical forecasting

The following benchmark concerns a different target and data design: historical next-day volatility rather than the preceding twelve-bar Bitcoin risk. Its scores are not pooled with the calendar-hour experiments.

14.1 How physics and forecasting are connected

The physics motivates observable activity, joint price–volume statistics, and extreme-concentration measures. Market tests assess the fixed interface and reveal temporal dependence. The forecasting application directly uses these observables and multiscale lags, without true network or cluster labels. The heterogeneous autoregressive (HAR) representation follows the multiscale approach of Corsi[8]. It is an empirical forecasting specification using mechanism-motivated observables and temporal lags, with forecast accuracy evaluated against named statistical references.

After the final five-minute bar of day d closes, the model forecasts the next UTC day:

$$RV_d = \sum_{j=1}^{288} r_{d,j}^2, \quad z_{d+1} = \log RV_{d+1}, \tag{39}$$

$$\widehat{z}_{d+1} = \bar{z}_{\text{fit}} + \widehat{w}^\top \widetilde{x}_d,$$
$$\widehat{w} = \arg\min_w \sum_{d \in \text{fit}} (z_{d+1} - \bar{z}_{\text{fit}} - w^\top \widetilde{x}_d)^2 + \alpha_R \|w\|^2. \tag{40}$$

Standardization uses the fitting period only. The ridge penalty α_R differs from a tail exponent. The displayed $100\sqrt{RV}$ measures intraday volatility intensity, without specifying the direction of the close-to-close return.

Table 10. The full model's ten inputs, all available at the forecast origin.

Group	Definition	Count
HAR	$\log RV_d$; log mean RV over the past 7 days; log mean RV over the past 28 days	3
Activity	Log total daily BTC volume; log ratio of daily volume to its past-7-day mean	2
Coupling	Bar-level Spearman$(\lvert r\rvert, v)$; $\overline{r_c^2 v_c}/(\overline{r_c^2}\,\bar{v})$; absolute summed signed-taker volume divided by daily volume	3
Tail	$\log[288 \sum r_j^4/(\sum r_j^2)^2]$; $\max r_j^2 / \sum r_j^2$	2

r_c, v_c are demeaned within a day. Signed activity derives from venue taker fields, not full order-book OFI or market-wide buy–sell imbalance. Tail inputs measure finite-sample shape and concentration, not power-law exponents. The 1/7/28-day horizons adapt the setup to a 24/7 market.

The original protocol used 337 eligible training targets in 2025 and 90 validation targets in 2026 Q1. After selection, one refit used all 427 targets. Parameters remained fixed across 183 evaluation dates from April 1 to September 30, 2026. Observed test-day data can become later lagged inputs, but do not enter refitting or renewed selection. Figure 20 retains every test date.

14.2 Overall skill and incremental value are separate

For D evaluation days, the primary loss and relative skill are

$$L_{\log} = D^{-1} \sum_d (\log RV_d - \widehat{z}_d)^2,$$
$$\text{Skill}_B = 100(1 - L_{\text{model}}/L_B). \tag{41}$$

Table 11 and Fig. 21 place simple and strong references together. The full model has positive historical skill over a constant, persistence, and EWMA. HAR alone already improves over persistence by 29.61%. The additional 2.263% point gain over HAR has an interval crossing zero, with Holm-adjusted $p = 0.3948$ in the original primary comparison. This contrast measures the combined addition of return-shape and trading-information features.

The output is a next-day volatility estimate paired with a date-by-date error record. Subsequent asset and time tests in Appendix D examine the transfer of the added features. Prospective calibration and decision-specific losses would be needed to assess operational risk use.

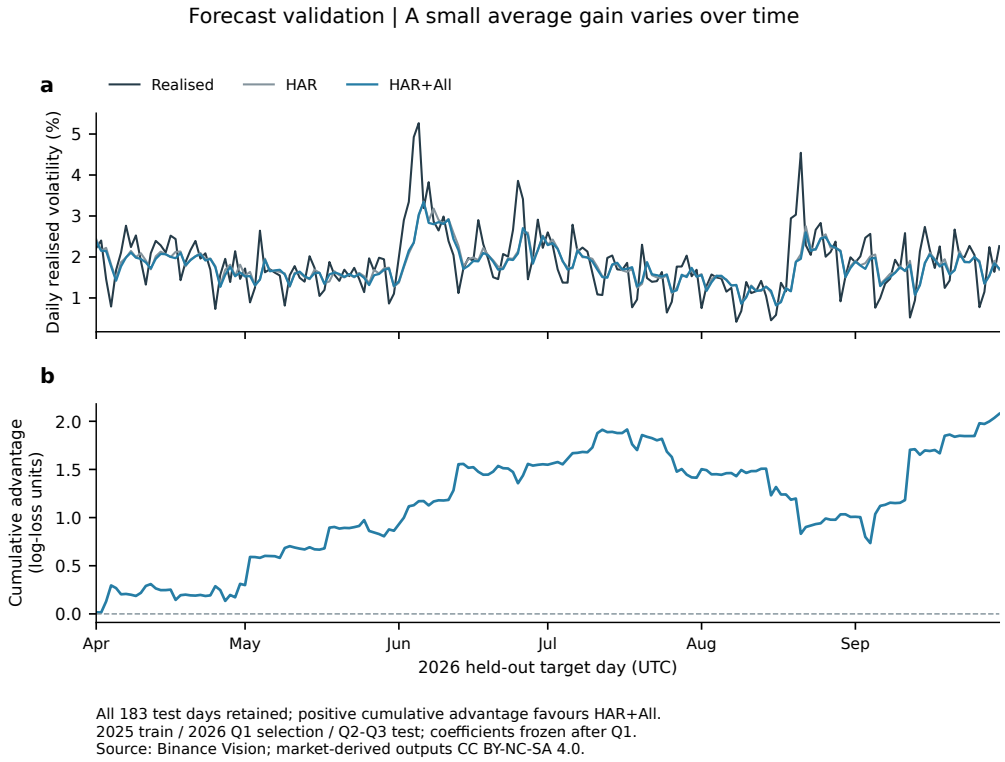


Figure 20. Forecasts and realizations on all 183 historical evaluation dates. Volatility is $100\sqrt{RV}$. The lower panel accumulates forecast-loss advantage over HAR. Binance Vision, CC BY-NC-SA 4.0. [P01]

Table 11. Original next-day log-RV evaluation: 183 BTCUSDT dates.

Reference/model	log-MSE	All loss reduction (%)	Approx. 95% CI (%)
Fit-period mean	0.793999	38.26	[24.65, 48.97]
Persistence	0.712561	31.20	[21.76, 38.62]
EWMA	0.606913	19.23	[13.22, 24.99]
HAR reference	0.501565	2.26	[-0.11, 4.56]
HAR+All	0.490213	—	—

All denotes the original HAR+All. Intervals use 4,999 paired circular seven-day blocks, unadjusted, in a post-analysis overall-skill reassessment. Source: predictive_utility_v1.

14.3 Mechanism-imposed risk under a matched information budget

The statistical feature benchmark above asks whether selected observables improve a HAR forecast. A separate mechanism test asks whether a pressure-state restriction adds information while preserving the same raw inputs and total-history budget. Both questions are retained, with their own protocols and scores.

The rolling mechanism study uses the same past price, volume, count and signed-flow fields as its macro-HAR reference. A reduced pressure model forecasts future flow moments and passes them through the response/risk map. At a 240-day total-history budget, its QLIKE is .260805; a same-map pressure AR control gives .260792, return-HAR .228191, and macro-HAR about .2236. The near identity with the pressure control means the tested network-motivated reduction supplies no demonstrated increment at this budget. The historical dates have been examined in prior project rounds and are development evidence, not a newly untouched test.

Further joint-flow-moment correction is evaluated at 30, 60, 120 and 240 days rather than selecting one favorable budget. At 240 days its QLIKE is .2749, versus .2236 for macro-HAR. Separately estimating residual cross moments produced incompatible covariance fits; the common-path construction in Section 10 addresses that calculation issue without retroactively changing these market scores.

A signed-flow/response bridge also generates sequential $\Delta = -9.562$, compared with -8.181 observed, while its selected additional forecast channel is Off. Its matched-HAR QLIKE .183724 and

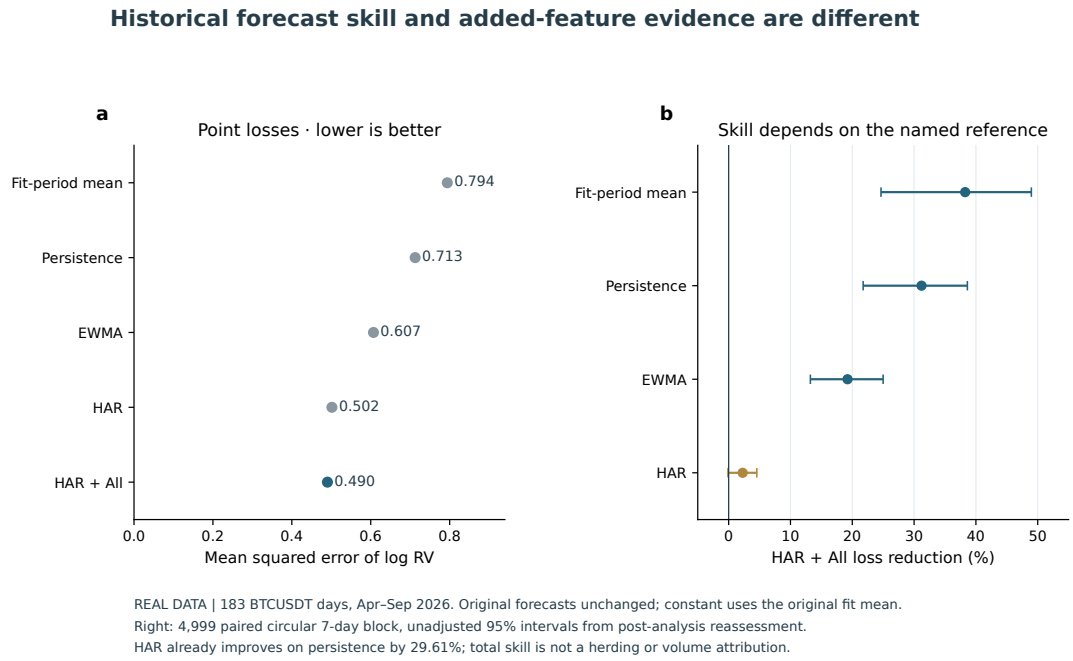


Figure 21. Historical log-MSE and full-model loss reductions against named references. Intervals use 4,999 paired circular seven-day-block resamples. The constant-reference comparison belongs to the post-analysis reassessment; the original HAR comparison is reported in the text. [J02]

selected shared hybrid .186232 belong to that different fixed protocol. Thus explaining a negative compatibility statistic and improving a next-day forecast are separately evaluated achievements. Scores from distinct protocols are not interpreted as a chronological improvement series.

Evidence record: matched-budget collective-dynamics scores, joint-flow W3 comparisons and the chronological signed-flow bridge benchmark.

15 Portfolio and strategy risk diagnostics

15.1 How much sample risk can common movement conceal?

The US-equity study contains nine sector ETFs, SPY and QQQ references, and 46,530 daily rows through October 23, 2026. Financial association networks have established precedents[9]; our sector graph represents observed co-movement, not trader interactions. The original design uses 2010–2017 for training, 2018–2019 for selection, 2020–2025 for historical evaluation, and 2026 for confirmation.

For within-window return covariance $\widehat{\Sigma}$ and weights w,

$$\widehat{v} = w^{\top}\widehat{\Sigma}w = \underbrace{\sum_{i} w_i^2\widehat{\sigma}_i^2}_{v_{\text{self}}} + \underbrace{2\sum_{i<j} w_i w_j \widehat{\sigma}_{ij}}_{v_{\text{collective}}}. \tag{42}$$

Annualized volatility is $100\sqrt{252\widehat{v}}$. For nine equally weighted sectors in the 63-day window ending March 31, 2020, the full estimate is 57.73%. Retaining only diagonal terms gives 20.62%, a ratio of 2.80. Collective terms account for 87.24% of full estimated variance. Both estimates use the same returns, weights and 63-day window; their variance difference isolates the sample covariance contribution. Figure 22 retains the longer rolling history.

The existing tool lets users change dates, 63/126-day windows, and holdings. It also evaluates $\Sigma_\eta = (1-\eta)\,\mathrm{diag}(\widehat{\Sigma}) + \eta\widehat{\Sigma}$, with $0 \le \eta \le 1$. The interpolation preserves positive semidefiniteness and gives a conditional covariance scenario for the selected holdings. The interface exposes how estimated common movement changes the same portfolio's risk.

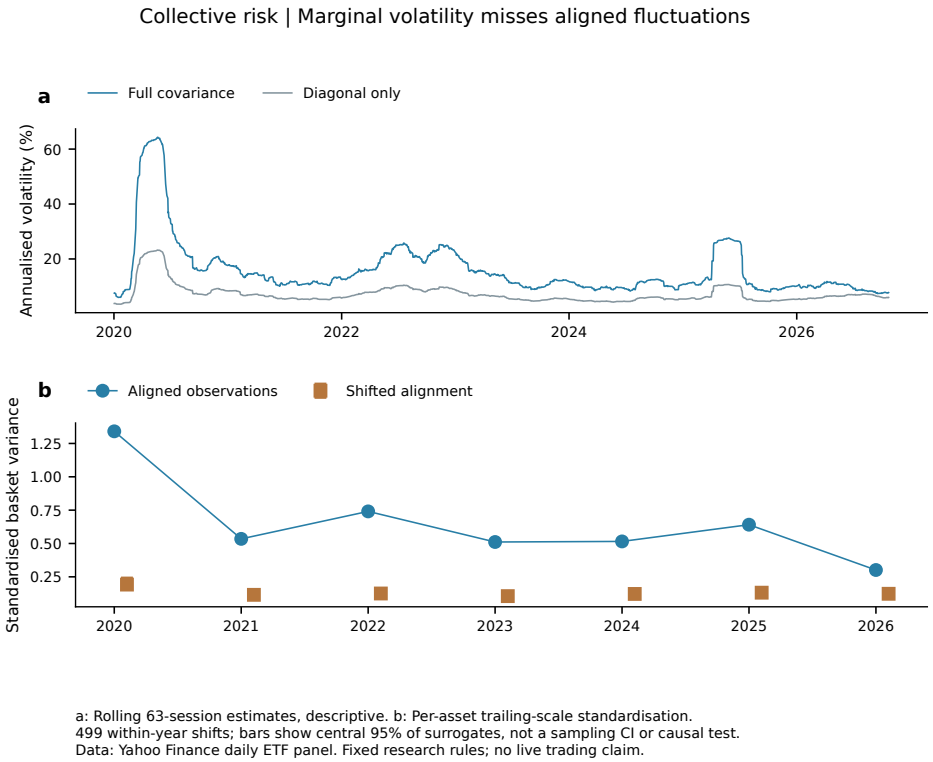


Figure 22. Sector-portfolio risk decomposition and conditional shifts of cross-asset date alignment. Numerical risk uses the full covariance matrix, independently of the plotted edge threshold. Shift ranges describe the conditional alignment comparison. The fixed 2020 case is in risk_lab_v1/risk_snapshots.csv. [U02]

15.2 Strategy variants are not independent risk sources

The six fixed proxies use momentum at 63/126 days, reversal at 5/21 days, and low volatility at 21/63 days. Dependence is assessed primarily through daily net P&L increments. The low-volatility pair has daily correlations 0.921/0.795 in the historical/2026 periods, versus cumulative-curve correlations around 0.982/0.985. Cumulative curves therefore cannot replace daily-risk analysis. Figure 23 separates shared exposures and cross-asset terms.

The default screening rule uses only 2011–2017 daily net increments and turnover. It ranks candidates by turnover and flags positive correlations ≥ 0.80. Six candidates become five representatives, reducing their count by 16.7%. Evaluation-period statistical reconstruction of the flagged strategy gives $R^2 = 83.9\%/64.1\%$. The rule was proposed after evaluation data had been inspected and remains exploratory. Reconstruction R^2 is neither retained alpha nor an executable hedge return.

Table 12. Default screening from six strategies to five representatives.

Period	Cumulative net P&L (%)	Annual P&L volatility (%)	Daily turnover (%)
2020–2025	-21.68→-20.77	4.35→4.33	7.46→8.67
2026 confirmation	-6.44→-5.57	3.84→3.99	8.03→9.05

Fixed reference capital with additive P&L; 5 bp one-way transaction cost and assumed 100 bp annual borrowing cost. Screening is post-analysis exploration, not a new blind test. Source: risk_lab_v1/subset_metrics.csv.

Table 12 shows higher combined turnover after removing one candidate, together with higher 2026 volatility. The useful workflow flags candidates, then checks combined holdings, costs, and exposures before deciding what to retain. The interface makes all 63 nonempty subsets available for comparing those consequences under a fixed evaluation period.

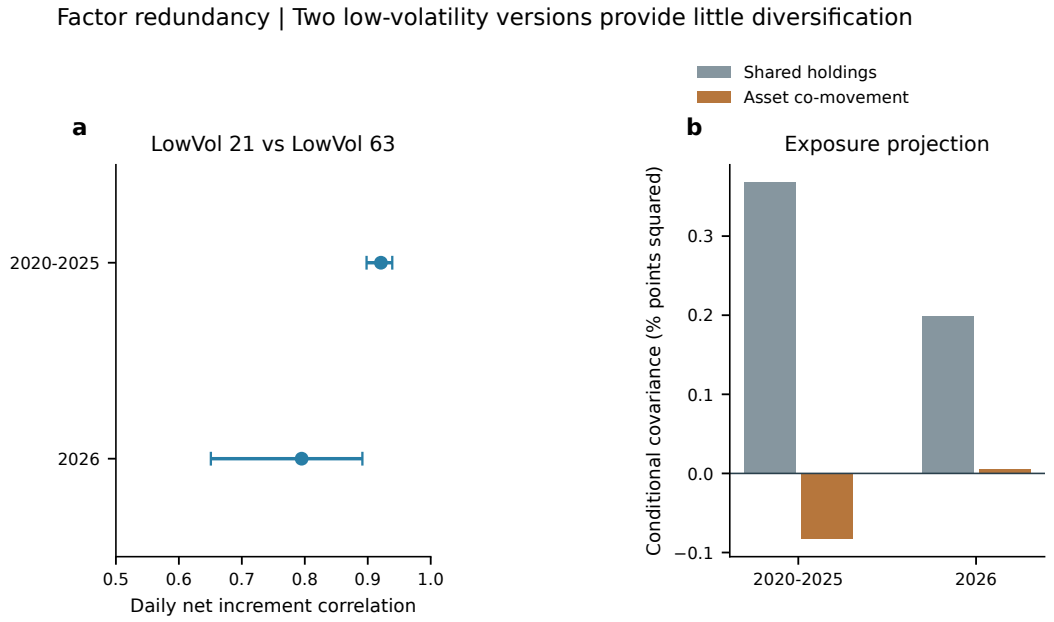


Figure 23. Daily net P&L dependence for the two low-volatility strategy variants, with cumulative-curve comparisons and exposure-based risk decomposition. Intervals use date blocks. For equal-risk strategies with daily P&L correlation c_{12}, $2/(1+c_{12})$ is a covariance-equivalent count. [U05]

15.3 How the mechanism enters the tool

With additional assumptions of common loading v, independent noise covariance D, and fixed strategy exposures W,

$$\Sigma_Z = Kvv^\top + D, \quad K = q(\lambda/N)^2 \mathbb{E}S_2,$$
$$\Sigma_{\text{strategy}} = W\Sigma_Z W^\top. \tag{43}$$

This conditional factor projection connects a coordination-generated common mode to the covariance used by the risk tool. Empirical covariance can also reflect common news, holdings, and heteroskedasticity. The projection describes fixed exposures; daily strategy P&L is evaluated separately with its actual time-varying holdings.

The existing single-file tool has four modules: omitted risk, factor screening, allocation comparison, and physical mechanisms. Its mechanism module varies ρ, q, redraws graphs, and simulates actions while displaying cluster moments and model risk. The common-loading scenario is an illustrative model calculation; sector risk uses the observed return covariance. Tool settings, source metrics, and implementation checks accompany the interface.

Source tables and methods: `web/advanced/index.html`.

16 Discussion

16.1 The useful object is a state-and-environment risk map

The finite reference supplies exact moments, recovery under known scales and a structural identification boundary. Effective weights retain its covariance mechanism while accommodating large decisions and fragmented execution. The interdependent network then supplies fast/slow memory, adaptation and accommodation, exact local moments and verified multistep propagation. The new calendar-time experiments complete the explanatory chain: common future decision innovations raise price risk conditionally, and lower capacity magnifies that increment even with identical volume, counts and pricing-noise paths.

This gives a concrete interpretation of shared quantitative-strategy exposure. Similar signals can generate aligned weighted trading innovations, while price recovery, return feedback and the receiving environment govern their risk transmission. The current scenario grid quantifies those channels without equating a factor label with observed trade alignment. It does not measure real fund portfolios, financing dependencies or regulatory welfare. Institutional observations and price-discovery or execution costs would be needed before turning this mechanism into a restriction on quantitative trading.

16.2 A recoverable mechanism effect is not necessarily a new predictor

The precision study finds real variation in covariance and feedback effects after reducing simulation noise. The information study locates a strong joint contribution from hidden activity, size, liquidity and concentration states. It also shows why adding a coordination-effect estimate can fail: that estimate may repeat information already summarized by activity and past volatility. The useful inverse target is therefore a distribution over effective trading and environmental structure that changes future risk, not a unique trader count or a scalar coordination score chosen in advance.

The empirical closure makes this target operational with the allowed macro history. Its internal point improvement over HAR remains statistically unresolved, and the exact local-noise cancellation shows how an ordinary return moment can masquerade as a distinct mechanism-derived forecast. Direct-moment controls are consequently part of the substantive result. Known source laws, simulated labels, derived features and calibration computations count toward the information budget; matched raw fields do not imply equal total resources. At present, predictive superiority, total information savings and reliable recovery of real-market latent structure have not been established.

16.3 Applications and the next discriminating observation

The portfolio, strategy-overlap and historical replay tools retain their original uses: expose common risk, distinguish observation from estimate, and let users inspect conditional consequences. The new Explore environment experiment adds a reproducible stress map with paired uncertainty. It lets a user vary common innovation, feedback, capacity and impact recovery, then compare the model scenario with the separate internal market forecast evidence. This is an explanatory risk interface rather than a live trading signal.

The next discriminating research step is to isolate which environmental component supplies the hidden-state gain, then test whether a low-cost observable accommodation proxy can recover it. Any added field must be supplied to the statistical reference, and the next performance comparison needs fixed new dates or another asset. Across-parameter calibration and the remaining activity/persistence mismatch must be carried into that interpretation. Continuing to improve only effect-label precision or to add redundant coordination scalars would not address the measured bottleneck.

Quantum models and quantum machine learning remain possible extensions for collective dynamics and inference. A suitable first task would compare observation needs, error and computational cost on the identifiable known-truth state problem. A common encoding cannot separate classical observation laws that are already identical; the choice therefore depends on a specific inferential or computational benefit[12], not an assumed advantage or a fixed implementation timetable.

17 Conclusion

The project links aggregate trading observations to effective statistical structure and future risk through one continuous mechanism. Exact finite-network mathematics establishes the reference inverse and its limits; weighted decisions, execution observations and temporal dependence make the interpretation operational. The retained interdependent network supports verified stochastic propagation and a conditional explanation of collective risk. New paired experiments show that the risk added by common innovations is larger under lower shock capacity and interacts with return

feedback. Independent future replications distinguish that average effect from its within-market forecasting value. An observable Bitcoin closure tests the connection fairly, while uncertainty, exact cancellation and direct-moment controls prevent a representation improvement from being misidentified as recovered coordination. The established contribution is a quantitative state-and-environment explanation of collective risk with executable tests of what macro data can reveal. Reliable incremental market prediction remains a separate, unresolved empirical requirement.

A Model context: adaptation, bubbles and crashes

The model families in Table 13 address complementary mechanisms. Cont–Bouchaud links random coordination and shared cluster actions to aggregate fluctuations[1]. Our finite implementation specifies both volume observation and unit-order execution. Lux–Marchesi adds switching between opinions and trader types, producing endogenous feedback and temporal structure[2]. The persistent-state experiment in Section 6 isolates temporal activity and response while retaining exogenous state dynamics.

Johansen–Ledoit–Sornette (JLS) relates reinforcing imitation to conditional bubble growth and stochastic crash risk[3]. Its simplified setup is

$$dp/p = \mu(t)\,dt - \kappa\,dj, \quad \mathbb{E}[dj] = h(t)\,dt, \quad \mu(t) = \kappa h(t). \tag{44}$$

The hazard $h(t)$ links growth conditional on survival to compensation for jump risk. The study discusses pre-crash patterns for Wall Street in 1929, 1962 and 1987, and Hong Kong in 1997. These published retrospective cases illustrate the bubble mechanism. The endpoint of conditional bubble growth and the random crash time are distinct model quantities.

Table 13. Mechanisms represented by three financial market models.

Model	Mechanism	Connection to the present analysis
Cont–Bouchaud[1]	Shared actions in random groups	Finite ER model, price–volume moments, inversion and compatibility tests
Lux–Marchesi[2]	Opinion and trader-type switching with profit feedback	Adaptive comparison for the temporal-state experiments
JLS[3]	Reinforcing imitation, hazard and conditional growth	Literature context for bubbles, crashes and criticality

B Complete tail and scaling objects

Cluster number density, node-weighted cluster size, and return distributions are different observables. Figures 24 and 25 retain cluster evidence and separate population size from temporal aggregation. Independent-model second-order aggregation differs from market memory.

Figure 26 adds the calibration-scaled market/model tail comparison to the joint-observation and memory diagnostics. All 28 market CSVs and original reproduction records are retained. Source maps distinguish model seeds from observable inputs.

C Exact checks and dependence controls

Theory files contain proofs of Eqs. (2)–(3), monotone identification, and chain/star and mixture equivalence. Rational enumeration and exact certificates check implementations. The original 480 theory checks and ten-study refit records are included. Small-graph enumeration and rational certificates provide independent computational checks of the stated identities. The marginal-preserving permutation comparison D03 isolates same-period coupling: shuffling volume preserves both empirical marginal distributions while changing their joint dependence.

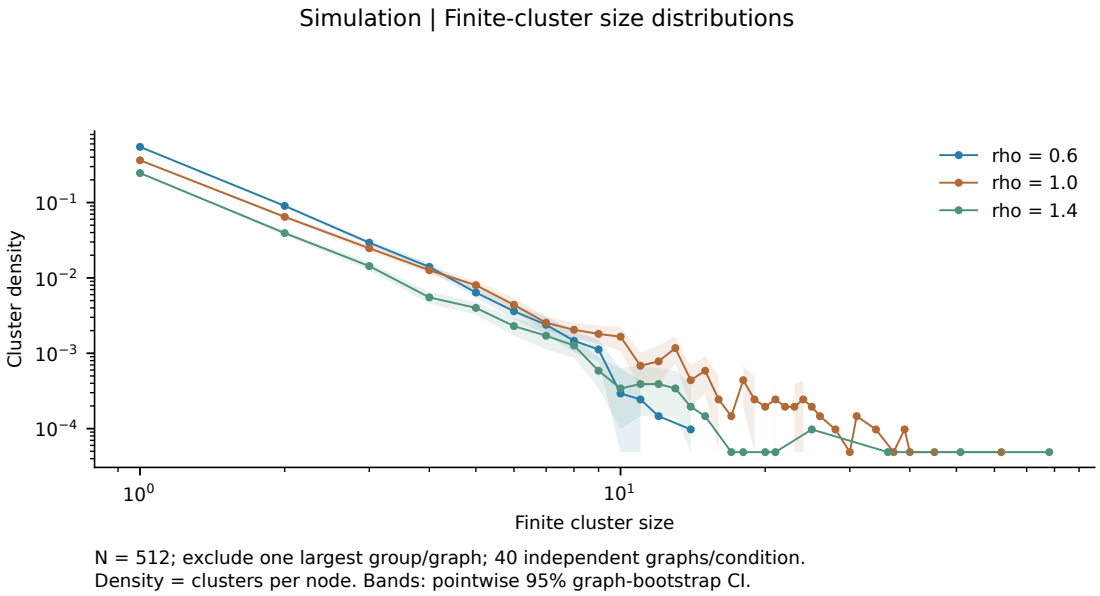


Figure 24. Finite-cluster number density at the specified population size. Pointwise bands resample graphs. Uniform-node sampling would weight each cluster by its size; raw cluster sizes and all population-size source tables are retained. [F04]

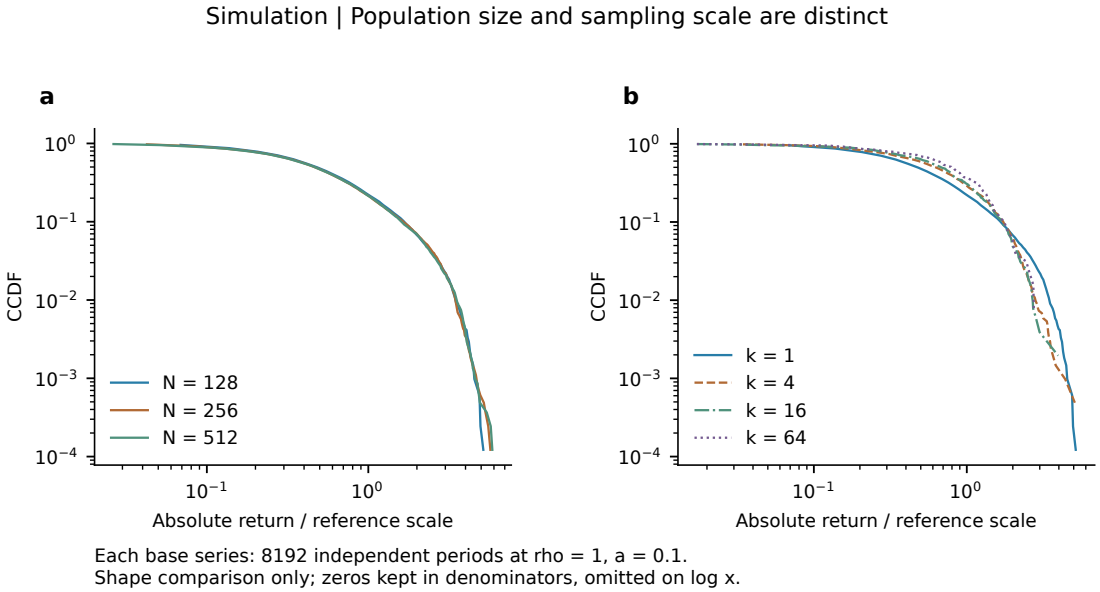


Figure 25. Separate comparisons of population size and temporal aggregation. Return standardization permits shape comparison; temporal blocks are nonoverlapping. The source tables retain all conditions. [F06]

D Supplementary forecasts and external evaluations

For a positive realized variance y and positive prediction μ, the quasi-likelihood loss (QLIKE) is

$$\ell_{\mathrm{QLIKE}}(y, \mu) = y/\mu - \log(y/\mu) - 1. \tag{45}$$

QLIKE is a secondary loss in the original log-RV benchmark and the primary loss in the subsequent state-forecast study. The latter's Gamma specifications use a log-linked positive mean and separate penalties on return and trading-information channels. Off denotes the matched return-only model with the trading-information channel fixed to zero. Feature and fitting definitions are in `docs/AB/state_forecast_v2_feature_spec.md`.

Table 14 retains supplementary losses. Monthly point gains over HAR are 2.90%, 5.41%, 4.99%, −1.12%, −1.71%, and 5.15%. A post-analysis comparison against HAR+Tail gives a 1.535% point

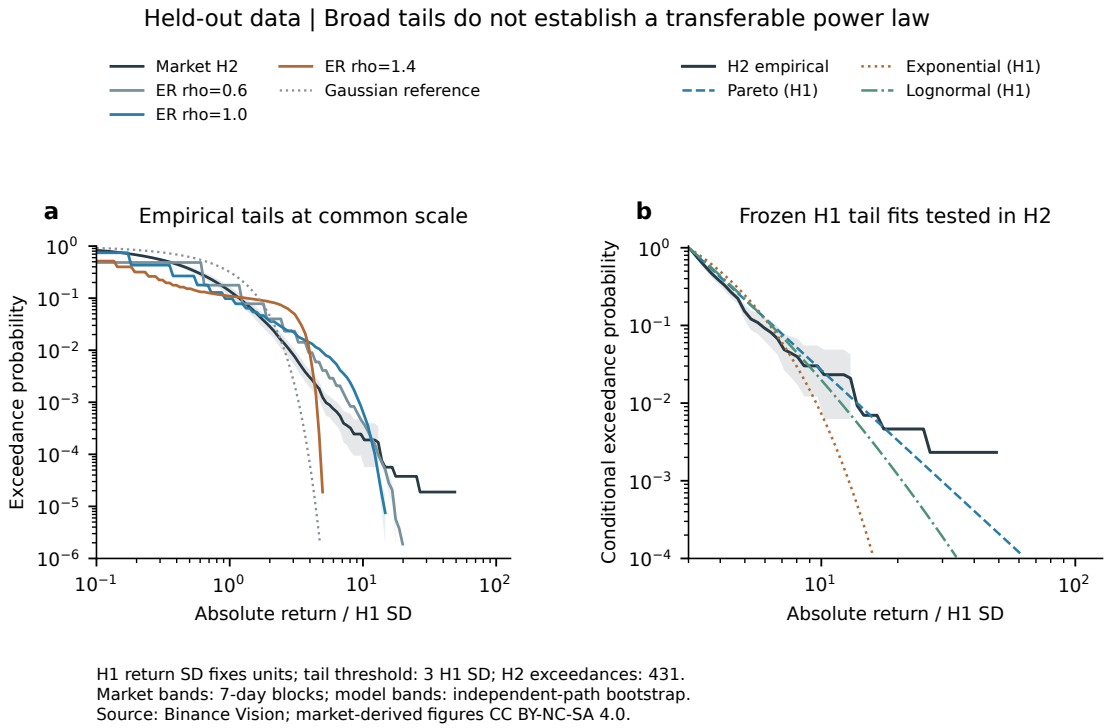


Figure 26. H1-scaled H2 market return CCDFs, conditional tails and continuous alternatives. Zero exceedances remain in the source data and are omitted from logarithmic display. Binance Vision, CC BY-NC-SA 4.0. [R02]

Table 14. Supplementary forecast losses, without replacing the primary metric.

Model	QLIKE	Volatility MAE (percentage points)
HAR	0.234214	0.440987
HAR+All	0.233933	0.433735

Volatility is $100\sqrt{RV}$. Exponentiating a log prediction need not yield a conditional arithmetic mean.

gain, again with an interval crossing zero. The final evaluation contains eight high-volatility events, giving a small event-specific assessment sample.

The subsequent state study adds an external ETH evaluation and 25 new dates for each of BTC and ETH. BTC's new-time primary QLIKE gain is 2.476%, with approximate 95% interval $[-0.407, 4.815]$%. Validation selects Off for the ETH trading channel. Complex Gamma specifications do not consistently beat simple HAR, as retained in Fig. 27. The BTC and ETH new-time evaluations cover the same 25 calendar dates.

On the same short BTC period, supplementary log-MSE improves by 3.170% against matched return-only Gamma. Its approximate 95% interval is $[1.236, 4.733]$%, with approximate Holm $p = 0.0276$ across that three-test supplementary family. This contrast assesses the trading-information channel within the Gamma specification. The primary assessment uses QLIKE and retains simple HAR among the references. Twenty-five dates provide only about 3.6 seven-day blocks, requiring a strong dependence approximation.

Covariance-matched tail tests hold information and covariance fixed, comparing nine methods on 38,070 rows from nine external stocks. Historical tail improvement is 0.557%, with Holm $p = 0.0772$; the 2026 point change is -1.797%. There are 32 historical co-crash events and zero in 2026, restricting the latter evaluation to false alarms. Figure 28 and the complete controls do not establish stable market increment. The external-stock evaluation tests transfer across assets over an already elapsed historical period.

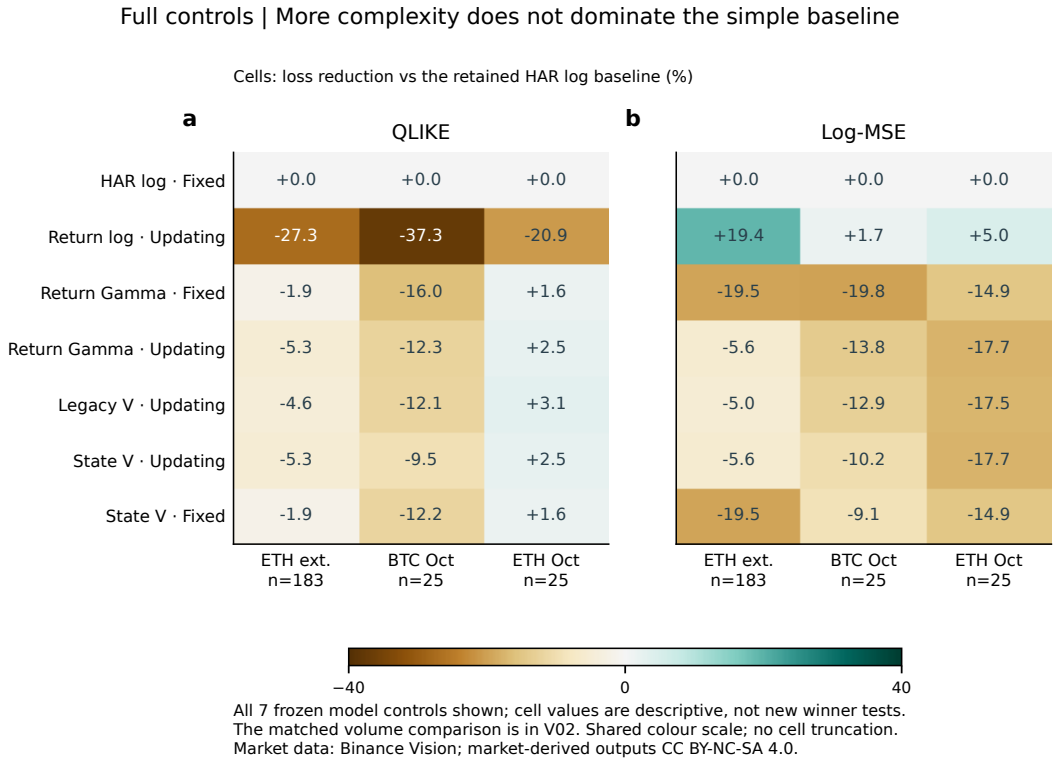


Figure 27. State/price–volume forecasts with simple HAR and matched controls. Previously inspected BTC periods are development information; external-asset and new-time evaluations are reported separately. Off denotes a validation-selected return-only fallback. Binance Vision, CC BY-NC-SA 4.0. [V03]

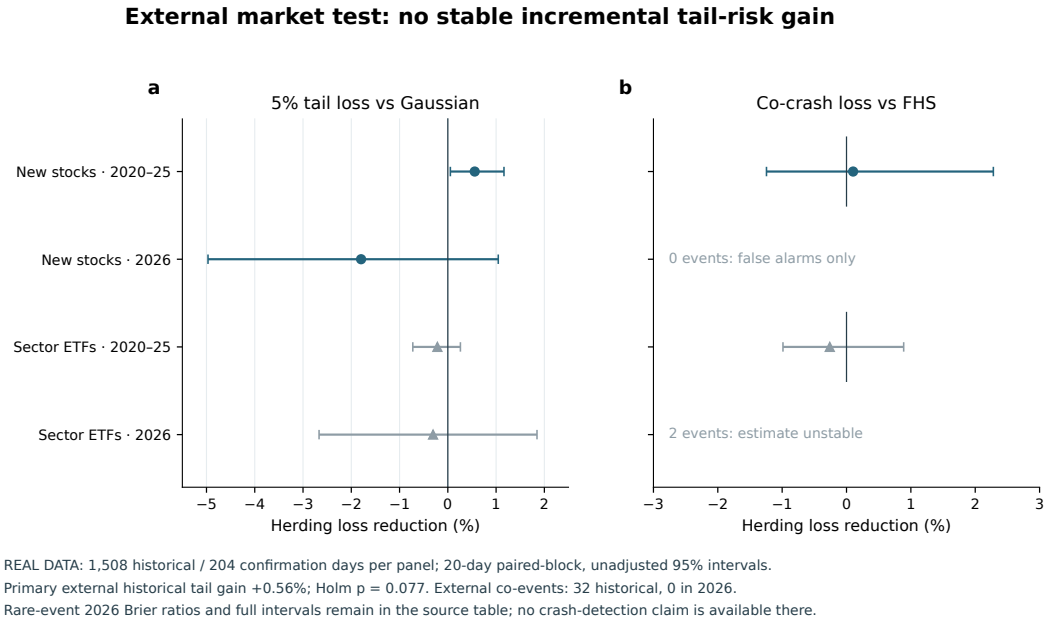


Figure 28. Tail and joint-downside comparisons at matched information and covariance. Statistical references, shrinkage, integration checks and criticality ablations are retained. No co-crash events occur in the 2026 sample, so that period evaluates false alarms. [T03]

E Stopped capacity extension and volume trade-offs

Six strategies, nine ETFs, 190 monthly snapshots, four allocations, and three depth levels produce 2,280 conditional states. Volume-only, one-round impact, and multiround-feedback capacity bounds coincide in every state, giving 0% extra reduction in Fig. 29. The maximum initial loss, 1.79158%, is below the fixed 2% trigger. A certificate also verifies nonactivation at smaller scales. The fixed

trigger is inactive throughout this design, which makes this branch a certificate of nonactivation under the tested self-impact scenario.

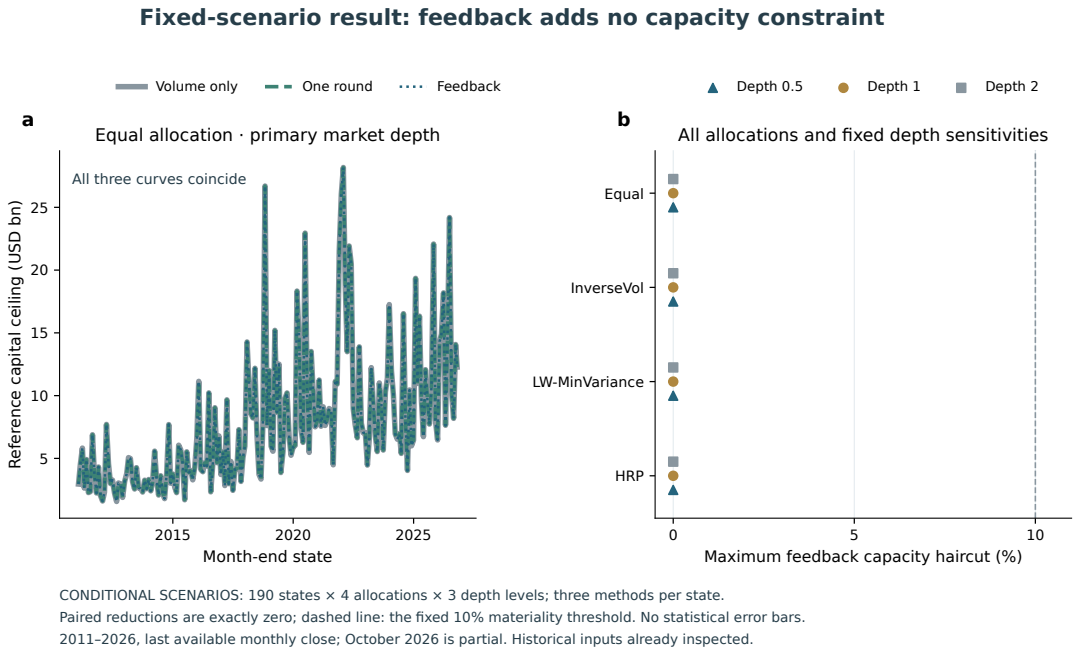


Figure 29. Coincident capacity bounds in the fixed account self-impact scenario. Historical prices and volumes supply conditional inputs; the feedback rule is specified rather than calibrated from execution data. All 2,280 states remain in the source record. [C01]

A conventional volume proxy still exposes a useful trade-off. Among 189 Equal-allocation historical replays with three complete future days, mean-volume sizing exceeds the participation proxy 142 times, or 75.13%. Using the past-20-day 20th-percentile dollar volume reduces this to 62 cases, or 32.80%, while mean deployed capital falls to 72.90%. These historical replays compare capital utilization with subsequent participation-proxy exceedances for a conventional volume rule, separately from the inactive feedback scenario.

The package includes full code, nine capacity CSVs, cash and monotonicity checks, C01–C03 plotting code, and the original certificate.

F Research record and reproduction

F.1 Finding the evidence

The public repository and research website link the shared implementation in `src/econophysics/`, configurations, numerical tables in `results/tables/`, source manifests and three executable notebooks. Bracketed figure identifiers retain the original F/D/R/P/U/V/T/C/J mapping; named continuation figures document the weighted dynamic extension. Table 15 lists the reproduction entry points.

F.2 Dynamic continuation evidence

The public package in `studies/collective_risk_continuation_v3/` contains numerical source tables, original hashes, protocols, independent arithmetic checks and the Explore data export. It distinguishes source-law simulations, the conditional-ensemble design and market-transport diagnostics. Primary ensemble splits are 2,048 bank, 32 tuning, 64 calibration and 128 evaluation trajectories. The earlier particle-model calibration is a separate design; its scores are not pooled with this evaluation.

Table 15. Reproduction profiles from the repository root. Each run writes new outputs.

Profile	Operation	Evidence
quick	Exact theory, model and source-table checks	Standard-library mathematics and numerical identities
theory	Exact identities and certificates	Calculations under the stated model assumptions
notebooks	Execute all three research notebooks	Model moments, inversion, market tests and forecast comparisons
site	Check local resources and the browser engine	Website links and agreement with Python fixtures
forecast	Download documented archives into a fresh directory and rebuild the fixed experiment	All 183 historical test dates and 16 saved CSV tables
Extended laboratory	Rebuild retained experiment pages	Risk, strategy overlap, allocation and coordination diagnostics

F.3 State–environment diagnosis and interactive evidence

The curated release in `studies/state_environment_v9/` retains fixed numerical tables, uncertainty results, verification records and provenance. It contains the 256-origin precision and information studies, the complete 29-cell response grid, paired environmental contrasts, all internal market scores and the post-run direct-moment controls. Figures prefixed `v9_` reproduce those source results. The independent validation record distinguishes arithmetic/implementation checks from scientific admission criteria.

The interactive entry is Explore: Risk environment, with the dedicated risk-environment page. Scenario estimates, latent capacity and intervention uncertainty remain visibly separate from observed Bitcoin outcomes. The downloadable state–environment evidence bundle supplies the fixed release tables. Earlier network, inversion, forecasting and portfolio evidence is retained alongside this continuation.

F.4 Environment and source records

Forecast reconstruction uses Python 3.14.4 and the versions in `requirements-study.txt`; plotting and notebook dependencies are listed separately. A fresh reconstruction reproduces all 16 saved forecast tables byte for byte. Notebooks retain executed outputs, and the quick and website checks pass on Python 3.11 and 3.14.

Binance Vision-derived results retain CC BY-NC-SA 4.0 attribution. Full supplier price histories are acquired through the source manifests. Code, data and third-party asset terms are documented in `LICENSE.md` and `DATA_LICENSES.md`. Historical experimental digests remain in the source record.

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